

Spectral Holography

Emergent Geometry, Gauge Theory, and Gravity from Modular Graph Entanglement

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Quantum gravity faces a trilemma: exact locality, a non-dynamical background, and unitary evolution cannot coexist. We resolve it by replacing spacetime with a finite weighted graph G whose vertices carry Type III₁ von Neumann algebras. The crossed-product construction promotes them to Type II_∞ factors, supplying a canonical trace and a modular clock. Self-consistency of the edge entanglement entropies gives a gap equation with a unique fixed point.

We approach this through a deliberate reframing of conformal compactification, AdS/CFT, and the string-theoretic origin of gauge fields. Penrose diagrams are read not as pictures but as combinatorial data: the integer N counting boundary components is the only conformal invariant, and we identify it with the number of extremal vertices of G . Standard holography (the Maldacena correspondence, the GKPW dictionary, the geodesic approximation to CFT correlators) and D-brane gauge dynamics are re-presented as limits of this graph structure. A numerical study of the geodesic approximation to the AdS₃/CFT₂ two-point function finds the semiclassical regime onset at $mR \gtrsim 5$, which we then identify analytically as the steepest-descent threshold of the graph heat-kernel resolvent.

From a single finite combinatorial object, the framework recovers a striking range of physics. The triangle graph C_3 with a $\mathbb{Z}_2$ holonomy yields exactly nine massless gauge bosons, the gluons and photon of the Standard Model, with no fine-tuning. The path graph P_2 exactly reproduces the eternal BTZ black hole and its known logarithmic entropy correction. The complete graph K_{10} matches the BFSS matrix model at quartic order, with maximal supersymmetry forced by Bott periodicity, identifying it as the M-theory saddle. A topological one-loop exponent $\alpha = (N_\partial + 1 + n_{\text{mod}} - n_{\text{gh}})/2$ controls logarithmic corrections to black-hole entropy on any graph: $\alpha = 3/2$ calibrates against BTZ, and $\alpha = 2$ on C_3 , $\alpha = 19$ on K_{10} are new parameter-free predictions. Assuming a Quantum Markov Property on cycles, we derive Newton's constant $G_N = 1/(4\lambda_1)$, the cosmological constant, the Ryu–Takayanagi formula, and the vacuum Einstein equations. The $K_{10} \rightarrow C_3$ transition amplitude is suppressed by M^{-16} , non-perturbative in the string coupling, suggesting structurally why perturbative string theory cannot describe the Big Bang.

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1	1	Introduction
2		Conformal compactification reduces any Lorentzian space-
4		time to a Penrose diagram with N boundary components;
5		up to conformal equivalence, N is the unique discrete in-
8		variant of the asymptotic causal structure. We propose
11		that the bulk is the spectral data of a finite weighted graph
21		$G = (V, E)$ whose N extremal vertices <i>are</i> the N bound-
		ary components. Edge weights $w_{ij} > 0$ carry bipartite
		entanglement; automorphisms $\Phi_{ij} \in \text{Aut}(\mathfrak{g})$ carry gauge
		holonomies. The bulk dimension is forced by N alone:
		spectral embedding sends each vertex to $\mathbf{x}_i \in \mathbb{R}^{N-1}$, the

angular polytope $\mathcal{P} = \text{conv}\{\mathbf{x}_i\}$ has dimension $N - 1$, and the warped bulk has $D_{\text{bulk}} = N + 1$ (one time, one radial, $N - 1$ angular). Known holographic examples are all consistent: $N=1 \rightarrow \text{AdS}_2/\text{SYK}$ [1–3]; $N=2 \rightarrow \text{BTZ}$ [4]; $N=10 \rightarrow 11\text{D M-theory}$. The graph hypothesis is therefore not merely compatible with holography in known cases; the bulk dimension is *predicted* from the only conformally invariant integer on the boundary.

The trilemma of quantum gravity [5] - exact locality, non-dynamical background, unitarity - is dissolved by all three becoming algebraic: locality is edge connectivity, background independence is the sum over graphs, unitarity is the canonical crossed-product trace.

Results are either unconditional, conditional on the Quantum Markov Property (Conj. 6.1), or phenomenological. Unconditional: the holonomy kernel theorem (Thm. 6.5) yields nine massless gauge bosons on $C_3/\mathbb{Z}_2$ without fine-tuning; the gap contraction lemma (Lem. 6.3) gives a unique fixed point in a Stieltjes cone whenever $\kappa < 1$; the path graph P_2 exactly reproduces eternal BTZ (§6.7) and Sen’s log correction $\alpha_{P_2} = 3/2$; the complete graph K_{10} at $\kappa = 0$ matches the bosonic BFSS Lagrangian at quartic order (Appendix A15) with $\mathcal{N} = 16$ from Cl(9) Bott periodicity; and the stratum invariant $\alpha = \frac{N\partial+1+n_{\text{mod}}-n_{\text{gh}}}{2}$ gives parameter-free predictions $\alpha_{C_3} = 2$ and $\alpha_{K_{10}} = 19$. Conditional on QMP: $G_N = 1/(4\lambda_1)$, $\Lambda_G = -N\lambda_1/[2(N-1)]$, the Ryu–Takayanagi formula, and the vacuum Einstein equations. The $K_{10} \rightarrow C_3$ trans-stratal amplitude is $O(M^{-16})$ by spectral flow – non-perturbative in $g_s = M^{-1}$, suggesting structurally why perturbative string theory cannot describe the Big Bang.

§6 develops the framework: crossed products, graph states, the gap equation, the holonomy kernel theorem, the spectral-superconformal algebra (SSCA), and the derivation of gravitational constants. §5.4 contains the numerical work establishing the saddle-point threshold $mR \gtrsim 5$. §6.7 and §6.8 treat the P_2 and K_{10} saddles. Background material on Penrose diagrams, Kaluza–Klein, D-branes, and the standard RT/island formulae appears in §2.1–5.3.

2 The discrete causal skeleton

2.1 Causal structure and the graph hypothesis

The causal structure of a Lorentzian manifold $(\mathcal{M}, g)$ is encoded in the light cone at each event: the null vectors $\{v \in T_p\mathcal{M} : g_{\mu\nu}v^\mu v^\nu = 0\}$ separate tangent space into timelike ($g(v, v) < 0$), null, and spacelike ($g(v, v) > 0$) sectors (Fig. 1) [6]. Causal curves have non-spacelike tangent everywhere. The causal future $J^+(S)$ (with respect to past $J^-(S)$) of a set S consists of points reachable by future-directed causal curves; an event horizon is $\mathcal{H} = \partial J^-(\mathcal{I}^+)$.

After conformal compactification $\tilde{g} = \Omega^2 g$, the asymptotic regions of $\mathcal{M}$ are mapped to a *finite* collection of boundary components $\partial\mathcal{M} = \{\Omega = 0\}$. These components are classified as future/past null infinity $\mathcal{I}^\pm$, time-like infinities $i^\pm$, and spacelike infinity i^0 . The Penrose

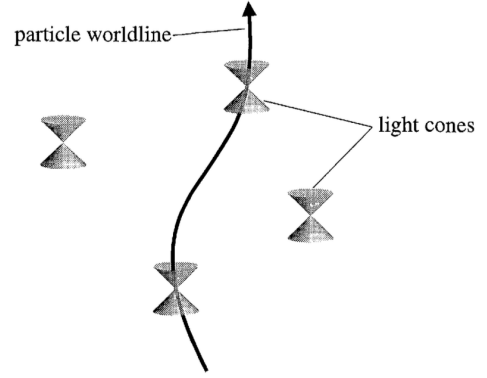


Figure 1: Light cones at a spacetime event: timelike directions lie inside the cone, null directions on the surface, and spacelike directions outside [6]. The future light cone consists of all future-directed null and timelike curves; the past light cone similarly. Events outside each other’s light cones are causally disconnected. This local causal structure is the seed for global causal relations such as horizons and the thermofield double state. In the discrete graph framework, the light cone structure is encoded in the spectral embedding metric and the edge connectivity.

diagram encodes exactly which of these components can send or receive causal signals to/from which others.

Remark 2.1 (The conformal invariant). The Penrose diagram is, considered up to conformal equivalence, characterised by a single discrete invariant: the integer N counting the connected components of $\partial\mathcal{M}$. Pure global AdS has $N=1$; the eternal BTZ black hole, the maximally extended Schwarzschild geometry, and the thermofield double all have $N=2$; multi-boundary wormholes have $N \geq 3$. Higher-dimensional data such as the conformal metric on each boundary and the asymptotic symmetry group are not invariants of the causal structure alone; only the integer N is.

The graph hypothesis presented here identifies N with the number of extremal vertices of a finite weighted graph G . The remaining algebraic data - entanglement weights w_{ij} on edges, holonomies $\Phi_{ij} \in \text{Aut}(\mathfrak{g})$ around cycles - has no direct correlate in the Penrose diagram; it is the bulk data the diagram does not see, supplied algebraically by the requirement of modular consistency (the gap equation, §6.1).

That entanglement entropy between two boundary regions is a physical observable depending only on the edge weight w_{ij} in G is consistent with algebraic QFT [7, 8], where causally disjoint regions generate commuting subalgebras and the net structure is determined by combinatorial data. The present construction differs from causal sets [8] in retaining the modular and gauge data on edges, not just causal relations on vertices.

The foundational claim of this paper, developed rigorously in §6, is that the finite graph G , with edge weights w_{ij} encoding bipartite entanglement entropies and edge

automorphisms Φ_{ij} encoding gauge holonomies, is the *fundamental object*. The bulk Lorentzian geometry emerges as a derived, large-scale phenomenon. The Penrose diagrams we construct below are not just pictures; they are the diagrammatic shadows of specific finite graphs.

The double-holography construction of Myers, Ruan and Ugajin [9] illustrates the picture sharply: two gravitating universes on codimension-one branes in a single asymptotically AdS_3 bulk with two disjoint boundaries, entangled in the thermofield double state

$$|\Psi\rangle = Z(\beta)^{-1/2} \sum_n e^{-\beta E_n/2} |E_n\rangle_L \otimes |E_n\rangle_R, \quad (1)$$

undergo a temperature-dependent topology transition between disconnected branes and a single connected brane (an Einstein–Rosen bridge). The Penrose-diagram invariant is $N = 2$, and the unique corresponding graph is the path P_2 with extremal vertices L, R joined by one edge. The absence of cycles enforces causal disconnection (QMP on trees, §6.1), the edge weight $w^* = S_{\text{BH}}$ encodes the entanglement, and the transition between the low-temperature disconnected phase ($w^* \rightarrow 0$) and the high-temperature ER-bridge phase (large w^*) is a dynamical instance of the island mechanism (§6.3.4). Taking the identification “ N boundary components = N extremal graph vertices” seriously drives every subsequent result.

2.2 Penrose diagrams for AdS and Schwarzschild

We construct Penrose diagrams for the two spacetimes central to holography and black hole physics: anti-de Sitter (AdS) and Schwarzschild, highlighting their distinct asymptotic boundaries and causal structures [10, 11].

2.2.1 AdS

d -dimensional anti-de Sitter space is the hyperboloid $-X_0^2 - X_{d+1}^2 + \sum_i X_i^2 = -R^2$ in $\mathbb{R}^{2,d-1}$. Global coordinates $(\tau, \rho, \Omega_{d-2})$ give

$$ds^2 = R^2(-\cosh^2 \rho d\tau^2 + d\rho^2 + \sinh^2 \rho d\Omega_{d-2}^2), \quad (2)$$

and the substitution $\tan \chi = \sinh \rho$, $\chi \in [0, \pi/2)$, brings the metric to a form conformal to the Einstein static universe with conformal boundary at $\chi = \pi/2$. The (τ, χ) Penrose diagram (Fig. 2) is a vertical strip; null geodesics reach $\chi = \pi/2$ in finite affine parameter and reflect, exhibiting the timelike nature of the AdS boundary [12]. Spacelike geodesics that connect to the boundary will be central to the bulk–boundary correlator analysis of §5.3.

2.2.2 Schwarzschild

The Schwarzschild metric $ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega_2^2$ with $f(r) = 1 - 2GM/r$ is brought to Kruskal–Szekeres form via the tortoise coordinate $r_* = r + 2GM \log|r/(2GM) - 1|$ and null coordinates $u = t - r_*$, $v = t + r_*$. Setting $U = -e^{-u/(4GM)}$, $V = e^{v/(4GM)}$ gives

$$ds^2 = -\frac{32(GM)^3}{r} e^{-r/(2GM)} dU dV + r^2 d\Omega_2^2, \quad (3)$$

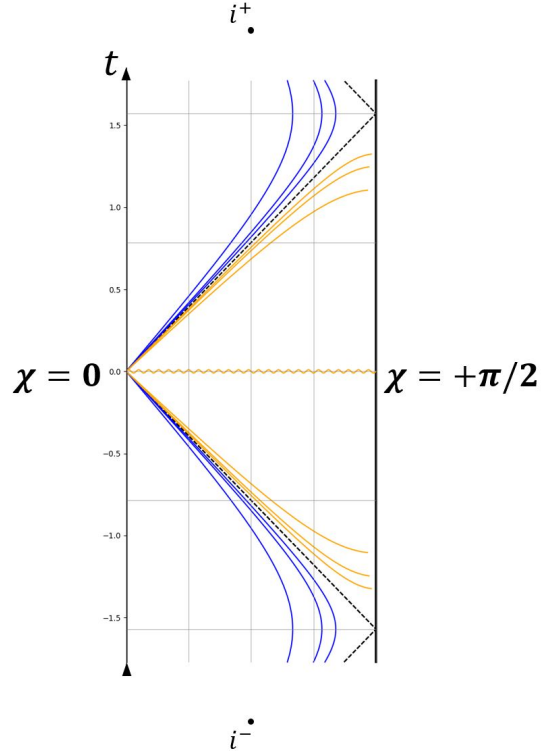


Figure 2: Penrose diagram of universal anti-de Sitter space (global AdS). The conformal boundary consists of the timelike surfaces $\mathcal{I}$ (shown as vertical line $\chi = +\pi/2$) and the disjoint points i^+ and i^- . Timelike geodesics are drawn in blue; they oscillate between the two timelike boundaries, reflecting the confining nature of AdS. Null geodesics (shown in a dotted black line) reach $\mathcal{I}$ in finite affine time and reflect. The timelike infinities $i^\pm$ are where a conformal field theory (CFT) can be defined; the AdS/CFT correspondence identifies the boundary CFT with the asymptotic data of the gravitating bulk.

with r defined implicitly by $UV = (1 - \frac{r}{2GM})e^{r/(2GM)}$. The full extension covers four regions: the physical singularity $r = 0$ at $UV = 1$, the event horizon at $UV = 0$, and two asymptotic exteriors. Compactify with $\tilde{U} = \arctan U$, $\tilde{V} = \arctan V$, and define

$$\tilde{T} = \frac{\tilde{V} + \tilde{U}}{2}, \quad \tilde{X} = \frac{\tilde{V} - \tilde{U}}{2}. \quad (4)$$

The resulting Penrose diagram (Fig. 3) is a diamond with the top and bottom corners cut off, representing the future and past singularities; the left and right boundaries are the two exterior null infinities, and the diagonal boundaries are the event horizons.

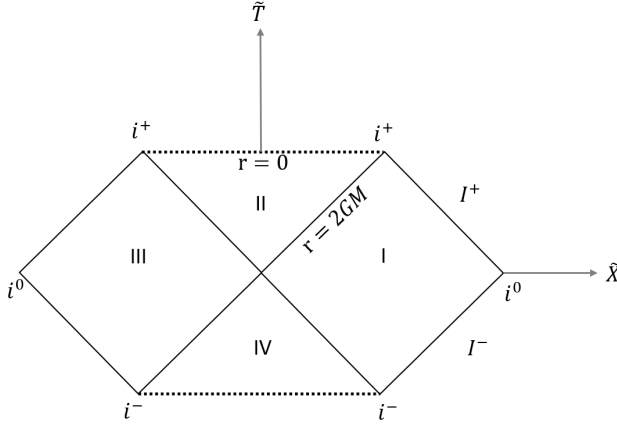


Figure 3: Penrose diagram of the global Schwarzschild black hole with singularities indicated by dotted lines and event horizons denoted by the diagonal lines crossing at the origin. The remaining diagonal lines bounding the left and right sides of the diagram are future and past null infinities I^+ and I^- respectively. i^+ , i^- and i^0 indicate future, past and spacelike infinity respectively. Region I represents outside of the black hole and II its inside. Similarly III is a second causally disconnected outside region and IV is the interior of a white hole.

3 CFT side

3.1 Curled up dimensions and Kaluza-Klein

The Kaluza-Klein (KK) mechanism is the cleanest realisation of the idea that 4D physics could emerge from a higher-dimensional theory [13]. Take five-dimensional spacetime $\mathcal{M}_4 \times S^1$ with the fifth coordinate $y \sim y + 2\pi r$. The signature of compactification is the appearance of infinite mass towers at $m_n = |n|/r$: every higher-dimensional field, on Fourier-expanding around the compact direction, gives a Kaluza-Klein tower of 4D modes. Lower bounds on the compactification radius from collider non-observation [14, 15] place $r \lesssim 10^{-19}$ m, but the mathematical mechanism remains central to every unification attempt. The

discrete graph version of this story, developed in §6, replaces the circle by a cycle graph and reproduces the same mass tower from the Laplacian spectrum.

3.1.1 Vector fields in 5D

Consider a 5D Abelian gauge field $A_M(x^N)$ on $\mathcal{M}_4 \times S^1$ with $y \sim y + 2\pi r$. The 5D Maxwell action is $S_{5D} = \int d^5x \frac{1}{g_5^2} F_{MN} F^{MN}$. Decompose $A_M = (A_\mu, A_4 \equiv \rho)$ and expand in Fourier modes:

$$A_\mu(x^\mu, y) = \sum_{n \in \mathbb{Z}} A_\mu^{(n)}(x^\mu) e^{iny/r}, \quad (5)$$

$$\rho(x^\mu, y) = \sum_{n \in \mathbb{Z}} \rho^{(n)}(x^\mu) e^{iny/r}. \quad (6)$$

Integrating over y yields the 4D effective action:

$$S_{4D} = \int d^4x \frac{2\pi r}{g_5^2} \left[\frac{1}{2} F_{\mu\nu}^{(0)} F^{(0)\mu\nu} + \frac{1}{2} \partial_\mu \rho^{(0)} \partial^\mu \rho^{(0)} + \sum_{n=1}^{\infty} \left(\frac{1}{2} |F_{\mu\nu}^{(n)}|^2 + \frac{n^2}{r^2} |A_\mu^{(n)}|^2 + \frac{1}{2} |\partial_\mu \rho^{(n)}|^2 + \frac{n^2}{r^2} |\rho^{(n)}|^2 \right) \right]. \quad (7)$$

with $F_{\mu\nu}^{(n)} = \partial_\mu A_\nu^{(n)} - \partial_\nu A_\mu^{(n)}$. The zero modes $A_\mu^{(0)}$ and $\rho^{(0)}$ are massless; the $n \neq 0$ modes have masses $m_n = |n|/r$. The 4D gauge coupling is $1/g_4^2 = 2\pi r/g_5^2$.

3.1.2 Physical implications and graph avatar

Compact extra dimensions modify Gauss's law at distances $\lesssim r$: $\Phi \propto 1/R^{D-3}$ crosses over to $1/R$ for $R \gg r$. Precision tests of Newton's law and collider searches for KK excitations constrain r to sub-TeV scales [14, 15]. For massless fields in D dimensions the little group $O(D-2)$ gives $(D-2)$ polarisations [16], which reorganise on compactification into the massive and massless 4D modes.

The KK spectrum $m_n = |n|/r$ on a circle is precisely the Laplacian spectrum of the cycle graph C_N in the continuum limit. Holonomy around the circle becomes a product of edge automorphisms; the massless condition $n=0$ corresponds to the fixed-point condition of Theorem 6.5. Non-Abelian gauge fields arise when $\mathcal{H} \subseteq \text{Aut}(\mathfrak{g})$ is non-Abelian: the twisted Laplacian kernel $\text{Fix}(\mathcal{H}|\rho)$ counts massless gauge bosons, and λ_1 sets the mass scale for the remaining modes.

3.2 SUSY Yang-Mills and non-Abelian gauge theories

A Yang-Mills (YM) connection A_μ^a valued in the Lie algebra of a non-Abelian gauge group H , with generators $[T_a, T_b] = f_{ab}^c T_c$ [17], transforms infinitesimally as

$$\delta A_\mu^a = (D_\mu \epsilon)^a = \partial_\mu \epsilon^a + g f_{bc}^a A_\mu^b \epsilon^c, \quad (8)$$

under which the field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + g[A_\mu, A_\nu]$, with $A_\mu = A_\mu^a T_a$, transforms covariantly, $F'_{\mu\nu} =$

$U^{-1}F_{\mu\nu}U$. For $H = SU(3)$ this is the gauge sector of QCD.

3.2.1 Supersymmetry

Quantum corrections introduce UV divergences that grow with energy; for the Standard Model the Higgs mass receives quadratically divergent contributions (the hierarchy problem). Supersymmetry softens these by enforcing cancellations between bosonic and fermionic loops, and maximally supersymmetric YM is conformal and finite - the natural setting for holographic dualities.

SUSY extends the Poincaré algebra by fermionic generators $Q_\alpha, \bar{Q}_{\dot{\alpha}}$ satisfying

$$\{Q_\alpha, \bar{Q}_{\dot{\alpha}}\} = 2\sigma_{\alpha\dot{\alpha}}^\mu P_\mu, \quad (9)$$

$$\{Q_\alpha, Q_\beta\} = \{\bar{Q}_{\dot{\alpha}}, \bar{Q}_{\dot{\beta}}\} = 0, \quad (10)$$

$$[P_\mu, Q_\alpha] = 0. \quad (11)$$

In a SUSY YM theory the gauge field acquires an adjoint gaugino λ^a ; the $\mathcal{N}=1$ action couples $F_{\mu\nu}F^{\mu\nu}$ to $\bar{\lambda}\sigma^\mu D_\mu\lambda$ with a topological $\theta F\tilde{F}$ term. Maximally supersymmetric $\mathcal{N}=4$ SYM in four dimensions, with gauge field A_μ , four Weyl fermions λ^I , and six adjoint scalars ϕ^i for gauge group $SU(N)$, is exactly conformally invariant - the property that links it to AdS_5 via the AdS/CFT correspondence and motivates the conformal-group analysis of §3.3. In the discrete framework of §6, the graph Dirac operator D_G plays the role of the supercharge Q and $\ker D_G = \text{Fix}(\mathcal{H}|_\rho) \cap \ker \hat{R}_P$ counts supersymmetric ground states combinatorially.

3.3 Conformal group

The conformal group preserves angles up to local rescaling, $g'_{\mu\nu}(x') = \Omega(x)g_{\mu\nu}(x)$ [17, 18]. For $d > 2$ it extends the Poincaré group by dilatations D and special conformal generators K_μ ,

$$[P_\mu, D] = iP_\mu, \quad [K_\mu, D] = -iK_\mu, \quad (12)$$

$$[P_\mu, K_\nu] = 2i(M_{\mu\nu} - \eta_{\mu\nu}D), \quad (13)$$

with the usual Lorentz brackets; the algebra is $SO(d, 2)$, the isometry of AdS_{d+1} [19]. For $d=2$ it becomes Virasoro and we work in the large-central-charge limit where $SO(2, 2)$ dominates. Primary operators satisfy $[K_\mu, \mathcal{O}(0)] = 0$ with dimension $\Delta \geq (d-2)/2$ for scalars.

The discrete framework realises this algebra in a strictly contracted form: only the Poincaré-dilatation subalgebra $\langle P_a, M_{ab}, D \rangle$ is exact at finite N . The special conformal generators $K_a = IP_a I$ require a conformal inversion that the bounded spectrum of the graph Laplacian forbids; equivalently, the angular polytope $\mathcal{P} \subset \mathbb{R}^{N-1}$ is piecewise flat with a single physical metric mode (Thm. 6.10), leaving K_a no angular degrees of freedom on which to act. Full $SO(d, 2)$ emerges only in the densification limit $N \rightarrow \infty$, where Weyl's law unfreezes a continuous tower of angular modes. The algebraic realisation of the unbroken generators, together with the supercharge D_G , is the spectral-superconformal algebra of §6.4.2.

4 Bulk

4.1 String actions

String theory considers the scenario in which the fundamental objects are not point-particles but relativistic strings, either closed up in loops or open. These strings have a tension which prevents them from being extended freely without an input of energy. One may write an action for these strings which minimises the area traced out by the string through spacetime (the worldsheet). An action which achieves this is the Nambu-Goto action [20, Chapter 6.4]:

$$S_{NG} = -\frac{1}{2\pi\alpha'} \int d\tau d\sigma \sqrt{-h}, \quad (14)$$

with the integrand being the infinitesimal area on the worldsheet, $h = \det h_{\alpha\beta}$ the determinant of the induced worldsheet metric

$$h_{\alpha\beta}(\tau, \sigma) = \partial_\alpha X^\mu \partial_\beta X^\nu g_{\mu\nu}(X), \quad (15)$$

$X^\mu(\tau, \sigma)$ the string position with worldsheet coordinates τ, σ and $g_{\mu\nu}$ the background metric. The string tension is $T = 1/2\pi\alpha'$.

One may also define the Polyakov action (in flat Minkowski) [20, Chapter 24.6]:

$$S_P = -\frac{1}{4\pi\alpha'} \int d\tau d\sigma \sqrt{-\gamma} \gamma^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu} \quad (16)$$

with $\gamma^{\alpha\beta}$ a new independent metric living on the worldsheet. Varying with respect to $\gamma^{\alpha\beta}$ and using (15) shows that $h_{\alpha\beta} \propto \gamma_{\alpha\beta}$, and substituting back recovers the Nambu-Goto action. The Polyakov action is thus the first order form of Nambu-Goto [17, Chapter 6] and is generally more useful with its additional Weyl invariance

$$\gamma'_{ab}(\sigma, \tau) = e^{2\omega(\sigma, \tau)} \gamma_{ab}(\sigma, \tau), \quad (17)$$

on top of the Poincaré and worldsheet diffeomorphism invariances already present in Nambu-Goto.

4.2 Open and closed strings

Strings may be open, with boundary conditions specified at their endpoints, or closed with periodic conditions. For closed strings, the Weyl and diffeomorphism invariances allow the conformal gauge $h_{ab} = \eta_{ab}$, in which the Polyakov action gives a 2-dimensional wave equation for X^μ with solution [20]

$$X^\mu = x^\mu + 2\alpha' p^\mu \tau + i\sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{1}{n} \left(\alpha_n^\mu e^{-2in(\tau-\sigma)} + \tilde{\alpha}_n^\mu e^{-2in(\tau+\sigma)} \right), \quad (18)$$

with x^μ the centre-of-mass position, p^μ its momentum, and $\alpha_n^\mu, \tilde{\alpha}_n^\mu$ the independent right and left oscillatory modes.

The constraint $T_{\alpha\beta} \equiv (2/\sqrt{\gamma})(\delta S_P/\delta \gamma^{\alpha\beta}) = 0$ from varying $\gamma^{\alpha\beta}$ defines the Virasoro generators $L_n, \tilde{L}_n$; promoting the oscillators to creation/annihilation operators

and defining number operators $N_n^\mu, \tilde{N}_n^\mu$ gives the closed-string mass spectrum [20]

$$M^2 = \frac{2}{\alpha'} \sum_{n,\mu} n (N_n^\mu + \tilde{N}_n^\mu) + \frac{4a}{\alpha'}, \quad (19)$$

with normal-ordering constant $a = -1$ for quantum consistency, which corresponds to the requirement of $D = 26$ dimensions for the theory [20].

The ground state $|0, 0; k\rangle$ has $M^2 = -4/\alpha'$, an unstable tachyonic state [20]. The first excited state $\alpha_{-1}^\mu \tilde{\alpha}_{-1}^\nu |0, 0; k\rangle$ is massless and rank-2, decomposing into a traceless symmetric part (the graviton $g_{\mu\nu}$), an antisymmetric traceless part $B_{\mu\nu}$, and a trace part (the dilaton ϕ) [17, Chapter 6]. A theory of gravity therefore emerges as a natural consequence of the closed string spectrum. Higher excitations give purely string-like vibrational modes with no point-particle analogue.

We will next see how massless vibrational modes of open strings, with ends anchored to lower-dimensional surfaces called D-branes, give rise to gauge fields confined to the surfaces of these D-branes.

4.3 Brane-world scenarios and the Karch-Randall setup

The Randall–Sundrum II (RS2) model [21] showed that an infinitely large extra dimension is compatible with effective 4D gravity, provided the warp factor $e^{-2|y|/\ell}$ traps the graviton zero mode near the brane. The metric is

$$ds^2 = e^{-2|y|/\ell} \eta_{\mu\nu} dx^\mu dx^\nu + dy^2, \quad (20)$$

with ℓ the AdS radius. The key insight is that branes can localise gravity even when the extra dimension is non-compact and infinite in extent. Karch and Randall [22] took this further, they asked what happens when the brane has tension σ and sits inside AdS satisfying the Israel junction conditions. For a 2-brane (codimension-1) in AdS_3 , the brane angle θ_B relates to tension by $\sigma = \sin \theta_B / \ell$. The induced Newton constant on the brane comes out as $G_N^{(2)} = G_N^{(3)} / (2\ell \sin \theta_B)$, which stays finite in the holographic limit. In the string-theoretic interpretation these are D-branes, providing a mechanism by which D-branes generate lower-dimensional gravity.

4.4 Gauge theories from D-branes

The combinatorial origin of non-Abelian gauge symmetry in string theory is a feature, not an accident: it falls out of the endpoint structure of open strings on coincident D-branes [20, 23]. For N coincident Dp -branes, open strings can start and end on any of them, giving N^2 sectors labelled $[ij]$. $i, j = 1, \dots, N$ tell you which branes the endpoints are stuck to (Figure 4). The massless spectrum consists of:

- $[ii]$ strings: gauge fields $A_\mu^{(i)}$ living on the i -th brane
- $[ij]$ with $i \neq j$: massive gauge fields when the branes aren't on top of each other

When the branes coincide, all N^2 strings go massless and you get a $U(N)$ gauge theory. The gauge bosons live in the adjoint, there are N^2 of them, and the low-energy effective action on the worldvolume is just the $U(N)$ Yang-Mills action. But $U(N)$ has a decoupled $U(1)$ factor, it's the centre-of-mass motion of the whole brane stack. For $N = 3$, take the state

$$|s_3\rangle = \frac{1}{\sqrt{3}} (|p^+, \vec{p}_T; [11]\rangle + |p^+, \vec{p}_T; [22]\rangle + |p^+, \vec{p}_T; [33]\rangle). \quad (21)$$

The corresponding gauge field $\bar{A}^{(3)}$ has charge $\bar{q}_3 = \frac{1}{\sqrt{3}}(q_1 + q_2 + q_3)$, which vanishes for all physical strings because $q_1 + q_2 + q_3 = 0$ always. The remaining $N^2 - 1$ gauge fields give you $SU(N)$ Yang-Mills for $N = 3$, QCD. This manifestly aesthetic structure, a non-Abelian gauge theory arising solely from coincident branes, is the combinatorial origin of the strong interactions.

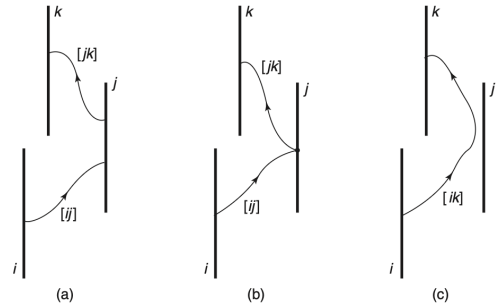


Figure 4: String joining on D-branes: (a) Strings in sectors $[ij]$ and $[jk]$ before interaction; (b) endpoints meet on brane j ; (c) strings join to form a single $[ik]$ string. This process generates non-Abelian interactions, giving $U(N)$ Yang-Mills theory for N coincident branes [20].

4.4.1 Gauge fields between separated D-branes

Separating the branes gives mass to the strings stretched between them. For two parallel Dp -branes separated by a distance d along a transverse direction, the mass spectrum becomes massive.

$$M^2 = \left(\frac{d}{2\pi\alpha'} \right)^2 + \frac{1}{\alpha'} (N^\perp - 1), \quad (22)$$

with $N^\perp$ the number operator. You get:

- Ground states ($N^\perp = 0$): massive scalars (tachyonic for small d , this is the instability)
- First excited states ($N^\perp = 1$): massive vectors and scalars

The massive vectors "eat" one scalar through the Higgs mechanism, becoming proper massive gauge fields with $D - 1$ degrees of freedom [20]. This is exactly how gauge bosons get masses in the Standard Model, just realised in string language. When branes have different dimensions,

you get ND coordinates: Neumann on one brane, Dirichlet on the other. These have half-integer mode expansions:

$$X^r(\tau, \sigma) = \bar{x}_2^r + i\sqrt{2\alpha'} \sum_{n \in \mathbb{Z}_{\text{odd}}} \frac{2}{n} \alpha_{n/2}^r e^{-in\tau/2} \cos \frac{n\sigma}{2}. \quad (23)$$

The normal-ordering constant picks up a contribution:

$$a_{\text{ND}} = \frac{1}{48}(p - q), \quad (24)$$

giving the mass formula:

$$M^2 = \left(\frac{\bar{x}_2^a - \bar{x}_1^a}{2\pi\alpha'} \right)^2 + \frac{1}{\alpha'} \left(N^\perp - 1 + \frac{p - q}{16} \right). \quad (25)$$

These sectors become crucial when you add supersymmetry as they produce chiral fermions, which are essential for building anything like the Standard Model [24].

Remark 4.1 (D-branes and SH). Each string-theoretic structure above has an algebraic counterpart in SH (§6): coincident D-branes $\rightarrow U(N)$ becomes Theorem 6.5; brane separation and Higgs mass generation becomes the spectral gap λ_1 ; gauge-gravity mixing via metric components is replaced by the Weitzenböck splitting $D_G^2 = \Delta_{\Phi, \rho} + \frac{1}{4} \hat{R}_{\mathcal{P}} + F_{\Phi}$ (Theorem 6.11), with off-diagonal coupling controlled by $\Xi_{aA} = [P_a, [T^A, D_G]]$.

4.5 Holographic entanglement and extremal surfaces

The Ryu–Takayanagi (RT) formula marks a decisive shift in perspective: the geometry of the bulk is not fundamental but emerges from the entanglement structure of the boundary. Entanglement entropy, originally a quantum information quantity defined solely by the reduced density matrix of a region, is here equated to a classical geometric object: the area of a minimal surface. This equivalence suggests that spacetime itself is a derived concept and given a boundary state with a specific entanglement pattern, one can reconstruct a bulk metric whose minimal surfaces reproduce the entanglement entropies of all boundary regions. The discrete graph framework of §6 will make this relation explicit: edge weights represent bipartite entanglement entropies, and minimal cuts in the graph become the discrete avatar of the Ryu–Takayanagi surface.

4.5.1 Ryu–Takayanagi formula

The RT formula [25] equates the entanglement entropy of a boundary region A with the area of a minimal bulk surface γ_A homologous to A :

$$S_{\text{EE}}(A) = \frac{\text{Area}(\gamma_A)}{4G_N}. \quad (26)$$

Subsequent work [26, 27] sharpened this into the statement that the connectivity of bulk geometry is a manifestation of boundary entanglement, with bulk distances controlled by entanglement strength.

4.5.2 Island Formula

Having established a direct dictionary between boundary entanglement entropy and bulk geometry, the quantum extremal island formula [28, 29] generalises this to include bulk quantum effects:

$$S(R) = \min \text{ext}_I \left[\frac{\text{Area}(\partial I)}{4G_N} + S_{\text{bulk}}(R \cup I) \right], \quad (27)$$

where I is an “island” region in the black hole interior. This formula resolves the information paradox by predicting a Page curve for Hawking radiation, reinforcing that spacetime connectivity is dual to quantum entanglement. In the discrete graph framework of §6, both the RT minimal surface and the island boundary ∂I become minimal cuts on a graph, with edge weights encoding entanglement entropies directly. The reformulation of RT in terms of bulk relative entropy [30] makes the modular-flow content of holographic entanglement manifest, a feature the graph framework inherits directly through the gap equation (52). The graph realisation of this formula, with islands identified as spectrally interior vertices and the Page curve emerging from gap-threshold crossings, is given in §6.3.4.

4.6 Junction conditions and brane dynamics

The embedding of a brane in a bulk spacetime is not arbitrary, it must satisfy the Israel junction conditions [31], which relate the discontinuity in extrinsic curvature across the brane to its stress-energy tensor. For a brane with tension σ embedded Z_2 -symmetrically in AdS_3 (meaning the bulk geometry is mirrored across the brane), these conditions reduce to

$$K_{\mu\nu} - K h_{\mu\nu} = -4\pi G_N^{(3)} \sigma h_{\mu\nu}, \quad (28)$$

where $h_{\mu\nu}$ is the induced metric on the brane, $K_{\mu\nu}$ is its extrinsic curvature, and $K = h^{\mu\nu} K_{\mu\nu}$ is its trace. Taking the trace gives $K = -8\pi G_N^{(3)} \sigma$, which determines how the brane bends the ambient geometry. These equations encode the backreaction of the brane on the bulk.

4.6.1 Induced brane cosmological constant

The geometry induced on the brane itself is governed by the Gauss–Codazzi equations, which relate the intrinsic curvature of the brane to the bulk Riemann tensor and the brane’s extrinsic curvature. For a codimension-one brane in AdS_3 , this yields the induced Ricci scalar

$${}^{(2)}R = -\frac{2}{\ell^2} + 32\pi^2 (G_N^{(3)})^2 \sigma^2, \quad (29)$$

where the first term comes from the bulk cosmological constant pulled back to the brane, and the second arises from the brane’s tension via the junction conditions. Rewriting this in terms of the brane cosmological constant $\Lambda_{\text{brane}} = {}^{(2)}R/2$, we obtain

$$\Lambda_{\text{brane}} = -\frac{1}{\ell^2} + \frac{\sin^2 \theta_B}{4\ell^2} \leq 0, \quad (30)$$

where we have used the relation between brane tension and brane angle $\sin \theta_B = \ell \sigma$, which follows from comparing the junction condition result $K = -8\pi G_N^{(3)} \sigma$ with the extrinsic curvature of a geodesic brane in global AdS_3 coordinates.

Crucially, Λ_{brane} is strictly negative for all physical brane angles $0 \leq \theta_B \leq \pi/2$, the $\sin^2 \theta_B$ term can never overcome the $-1/\ell^2$ contribution. This is the Karch-Randall obstruction [22]: a stable Karch-Randall brane in AdS_3 always carries negative induced cosmological constant. Obtaining a positive Λ (de Sitter space) from such constructions would require either complexifying the brane angle or moving beyond the simplest brane setups, both of which take us outside the regime where the Israel conditions as written here apply. This limitation is not just a technical inconvenience, it reflects an obstacle to realising accelerated expansion from brane-world scenarios in their most straightforward form.

4.7 p-branes are D-branes

Long before anyone understood D-branes as the surfaces where open strings end, supergravity already had an answer to the question of what sources the Ramond–Ramond (RR) gauge fields. That answer was the *p-brane*: a classical solitonic solution of Type II supergravity, generalising extremal black holes to objects extended in p spatial dimensions [32, 33]. The metric of an extremal p -brane in ten dimensions looks like

$$ds^2 = H(r)^{-1/2} \eta_{\mu\nu} dx^\mu dx^\nu + H(r)^{1/2} (dr^2 + r^2 d\Omega_{8-p}^2), \quad (31)$$

with $H(r) = 1 + (r_0/r)^{7-p}$. What makes the extremal case special is that it saturates the BPS bound: the mass per unit p -volume equals the charge density under the $(p+1)$ -form RR gauge potential C_{p+1} . Consequently, the near-horizon region becomes an $\text{AdS}_{p+2} \times S^{8-p}$ throat rather than a singular black hole, and the solution preserves half of the spacetime supersymmetries [34]. These were elegant geometries, but their microscopic origin was a complete mystery.

For years, the supergravity p -branes remained purely classical curiosities. The fundamental string couples to the Neveu–Schwarz two-form $B_{\mu\nu}$, not to the RR fields, so perturbative strings could not explain who was actually carrying the RR charge. This changed with Polchinski’s 1995 identification [23]: the D-branes we met earlier as boundary conditions for open strings are precisely the objects that source RR charge. A D p -brane couples minimally to C_{p+1} through the Wess–Zumino term in the Dirac–Born–Infeld action:

$$S_{\text{DBI}} = -T_p \int d^{p+1} \xi e^{-\phi} \sqrt{-\det(g_{ab} + B_{ab} + 2\pi\alpha' F_{ab})} + T_p \int C_{p+1}, \quad (32)$$

where $T_p = 1/[(2\pi)^p \alpha'^{(p+1)/2} g_s]$ is the brane tension [35]. The punchline: this tension matches exactly the ADM

mass per unit volume extracted from the supergravity p -brane metric. The two descriptions – a classical soliton of supergravity and a surface where open strings end – are two faces of the same coin.

The logic has a beautiful duality. At weak coupling $g_s \ll 1$, the D-brane is best described by its open-string degrees of freedom: a thin defect carrying a gauge theory on its worldvolume. At strong coupling $g_s \gg 1$, the brane backreacts heavily on the geometry and is more conveniently seen as the supergravity solution. These are complementary windows, and the BPS condition guarantees that protected quantities, like the charge and tension, can be safely compared across the two regimes without worrying about quantum corrections [23, 36].

It is exactly this equivalence that Maldacena exploited to formulate the AdS/CFT correspondence [12]. Take a stack of N coincident D3-branes. Look at it from the open-string perspective: in the decoupling limit you get $\mathcal{N} = 4$ SYM. Look at it from the closed-string perspective: the near-horizon geometry becomes $\text{AdS}_5 \times S^5$. The realisation that these two descriptions must be equivalent is the seed of the most powerful holographic duality we know. So the p -brane is the D-brane; the black hole is the gauge theory.

4.8 M-theory and BFSS: preview

M-theory is the conjectured $D=11$ quantum theory underlying Type IIA at strong coupling [36]; its low-energy limit is 11D supergravity (32 real supercharges). In the infinite-momentum frame it is conjectured to be the BFSS matrix model [37, 38],

$$S_{\text{BFSS}} = \frac{1}{g_s} \int dt \text{Tr} \left[\frac{1}{2} (D_t X^I)^2 + \frac{1}{4} [X^I, X^J]^2 + \text{ferm.} \right], \quad (33)$$

with $I=1, \dots, 9$ and $g_s=N^{-1}$, requiring $N \rightarrow \infty$.

SH selects K_{10} as the unique M-theory saddle via three algebraic facts derived in §6.8: $\text{Cl}(N-1)$ at $N=10$ gives a real Majorana spinor of dimension 16 (Bott periodicity); tower termination forces $D_{\text{bulk}}=N+1=11$ (Appendix A14); gravitational constants match 11D supergravity. Expanding the spectral action around K_{10} in the large- M limit (trace rank, $g_s=M^{-1}$) reproduces BFSS to quartic order (Appendix A15). The vertex count N and trace rank M play distinct roles; the ensemble sums over N .

5 AdS/CFT and holography

5.1 Definition

Consider N coincident D3-branes in Type IIB string theory. The total action splits as $S = S_{\text{bulk}} + S_{\text{brane}} + S_{\text{int}}$, where S_{bulk} describes closed strings, S_{brane} open strings, and $S_{\text{int}} \propto \kappa \sim g_s \alpha'^2$ their interactions [17, 19]. In the low-energy limit $\alpha' \rightarrow 0$, $S_{\text{int}} \rightarrow 0$ and massive string modes decouple, leaving only massless modes. The bulk reduces to free $D=10$ supergravity, while the open-string sector on the D3-branes becomes $\mathcal{N} = 4$ $SU(N)$ super

Yang–Mills theory in $3 + 1$ dimensions. Thus we obtain a decoupled system: free gravity plus a 4-dimensional CFT.

Now replace the D3-branes by their gravitational back-reaction: the extremal p -brane solution (31) with $p = 3$ gives

$$ds^2 = H(r)^{-1/2} \eta_{\mu\nu} dx^\mu dx^\nu + H(r)^{1/2} (dr^2 + r^2 d\Omega_5^2), \quad (34)$$

$$H(r) = 1 + \left(\frac{R}{r}\right)^4. \quad (35)$$

For an observer at infinity, energy E is redshifted relative to local energy E_p at radius r : $E = H(r)^{-1/4} E_p$. In the $\alpha' \rightarrow 0$ limit, two low-energy regimes decouple:

- Far from the brane ($r \rightarrow \infty$): the geometry becomes flat 10-dimensional Minkowski space, giving free supergravity.
- Near the horizon ($r \ll R$): $H(r) \approx (R/r)^4$, and arbitrary local excitations E_p appear as low-energy modes at infinity because $H^{-1/4} \rightarrow 0$.

These two regions decouple because climbing out of the gravitational well becomes impossible in the limit. Hence we obtain a free supergravity theory in flat space decoupled from a gravitational theory in the curved near-horizon geometry.

Comparing with the D-brane analysis, the free supergravity sectors match. The remaining sectors must therefore be equivalent: $\mathcal{N} = 4$ SYM on the D3-branes must correspond to the near-horizon supergravity. Taking $r \ll R$ in the p -brane metric yields

$$ds^2 = \frac{r^2}{R^2} \eta_{\mu\nu} dx^\mu dx^\nu + \frac{R^2}{r^2} dr^2 + R^2 d\Omega_5^2, \quad (36)$$

which is precisely $\text{AdS}_5 \times S^5$ with AdS radius R . Introducing $x_0 = R^2/r$ gives the standard form

$$ds^2 = R^2 \frac{\eta_{\mu\nu} dx^\mu dx^\nu + dx_0^2}{x_0^2} + R^2 d\Omega_5^2, \quad (37)$$

where the CFT lives on the conformal boundary $x_0 \rightarrow 0$, i.e., 4-dimensional Minkowski space. Thus the duality is established: $\mathcal{N} = 4$ $SU(N)$ SYM is holographically dual to Type IIB supergravity on $\text{AdS}_5 \times S^5$ [12].

5.2 AdS/CFT dictionary

There is an established dictionary for the AdS/CFT correspondence, mapping objects in the gravity theory on the bulk AdS side to objects in the gauge theory on the boundary CFT side. Specifically bulk fields $\phi(\vec{x}, z)$ with mass m are dual to operators $\mathcal{O}(\vec{x})$ of $\mathcal{N} = 4$ SYM with conformal dimension Δ , which is related to the mass by,

$$\Delta = \frac{d}{2} + \sqrt{\frac{d^2}{4} + m^2 R^2}, \quad (38)$$

with d being the dimensionality of the boundary of AdS_{d+1} (analogous formulae hold for higher-spin fields). The relation between bulk fields and CFT operators is explicitly displayed in the GKPW prescription [12, 17, 19]:

$$\left\langle \exp \left(\int d^4 x \phi_0(\vec{x}) \mathcal{O}(\vec{x}) \right) \right\rangle_{\text{CFT}} = Z_{\text{string}}[\phi_0(\vec{x})] \quad (39)$$

where $\phi_0(\vec{x})$ is the value the bulk field takes at the boundary (i.e. $\phi_0(\vec{x}) = \phi(\vec{x}, z)|_{z=0}$). The CFT part on the left-hand-side of the prescription is the generating functional of correlators in the field theory and on the right-hand-side is the string theory partition functional concerning a gravitational bulk field, specifically evaluated at the boundary. Notice the coupling between ϕ_0 and $\mathcal{O}$ in the CFT generating functional, which can be interpreted as such: bulk fields at the boundary ($z \rightarrow 0$) appear to act as sources for CFT operators living on the boundary.

5.3 Bulk Geodesics and CFT Correlators

A beautiful result of AdS/CFT, explicitly linking features of bulk geometry to boundary observables, is that the length of spacelike geodesics $\ell(x_1, x_2)$ connecting boundary points x_1, x_2 approximates two-point correlators of the dual CFT [39]:

$$\langle \mathcal{O}(x_1) \mathcal{O}(x_2) \rangle \sim e^{-m\ell(x_1, x_2)}, \quad (40)$$

where m is the mass of the bulk field ϕ dual to $\mathcal{O}$. The approximation is valid in the semiclassical limit $mR \gg 1$.

To investigate it explicitly, consider $\text{AdS}_3/\text{CFT}_2$. The metric in Poincaré coordinates is

$$ds^2 = \frac{R^2}{z^2} (-dt^2 + dw^2 + dz^2), \quad (41)$$

with curvature R and radial coordinate z . The boundary is at $z \rightarrow 0$, so points are $x_i = (t_i, w_i)$ and the CFT dimension is $\Delta = 1 + \sqrt{1 + m^2 R^2}$.

On a constant-time slice ($dt = 0$), the spacelike geodesic action is

$$S_{\text{geo}} = R \int \sqrt{\frac{\dot{w}^2 + \dot{z}^2}{z^2}} d\lambda, \quad (42)$$

with affine parameter λ . Translation invariance in w gives conserved momentum $\dot{w}(z) = Pz^2$. The affine normalisation $g_{ij} \dot{x}^i \dot{x}^j = 1$ yields $\dot{w}^2 + \dot{z}^2 = z^2$, and substituting $\dot{w}$ gives $\dot{z}^2(z) = z^2(1 - P^2 z^2)$. Integrating leads to the circle equation

$$(w - C)^2 + z^2 = \frac{1}{P^2}, \quad (43)$$

a semicircle with centre on the boundary ($z = 0$) and radius $r = 1/P$. The geodesic length, regularised by a cutoff $z = \delta \ll 1$ near the boundary, is

$$\ell = 2R \ln \frac{2r}{\delta} = 2R \ln \frac{|x_1 - x_2|}{\delta}, \quad (44)$$

because $|x_1 - x_2| = 2r$. Hence

$$e^{-m\ell(x_1, x_2)} = \left(\frac{\delta}{|x_1 - x_2|} \right)^{2mR}. \quad (45)$$

Comparing with the exact two-point function $\langle \mathcal{O}(x_1) \mathcal{O}(x_2) \rangle \sim |x_1 - x_2|^{-2\Delta}$, the geodesic approximation holds in the classical limit $mR \gg 1$ (where $\Delta \approx mR$):

$$\langle \mathcal{O}(x_1) \mathcal{O}(x_2) \rangle \sim \lim_{z \rightarrow 0} z^{-2mR} e^{-m\ell(x_1, x_2)}. \quad (46)$$

5.4 Numerical analysis of the Correlator/Geodesic approximation

To quantify the semiclassical regime, we computed exact and approximated correlators for varying mR (setting $R = 1$). Geodesics were found numerically via a shooting method: integrating the geodesic equations from the turning point z^* (where $\dot{z} = 0$) outward to $z = \delta$, using the affine normalisation $\dot{w}^2 + \dot{z}^2 = z^2$. The numerical length was verified against the analytical expression (44) for $|x_1 - x_2| = 2$, $\delta = 10^{-3}$, giving a relative error $\sim 10^{-8}$ (Figure 5).

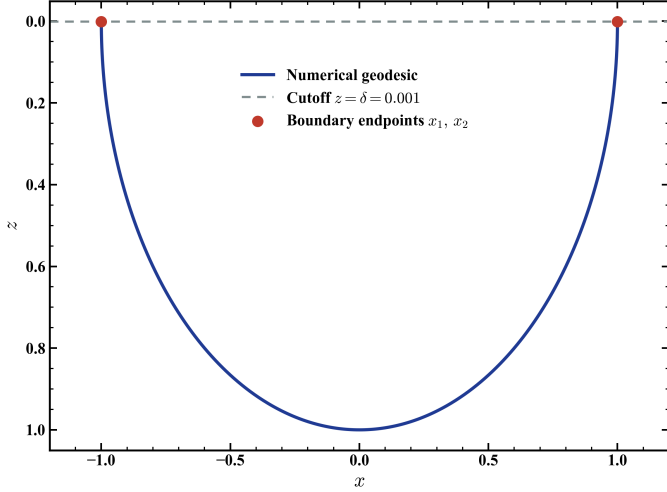


Figure 5: Numerically computed spacelike geodesic for AdS_3 connecting boundary points x_1 and x_2 . The semi-circular shape matches Eq. (43).

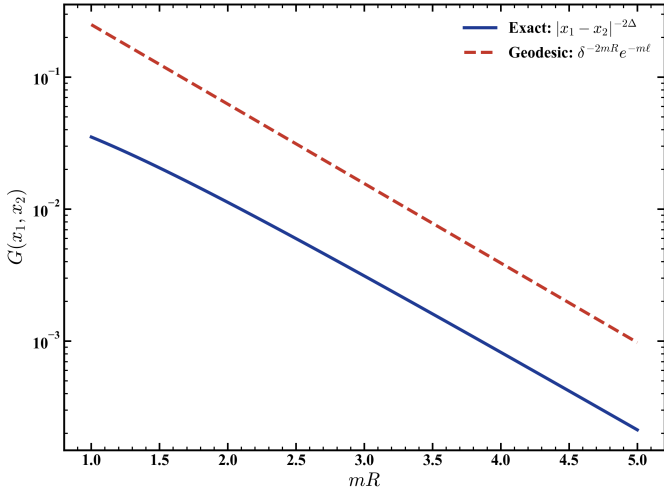


Figure 6: Exact correlator $|x_1 - x_2|^{-2\Delta}$ and geodesic approximation $e^{-m\ell}$ for varying mR .

The exact and approximate correlators are plotted side-by-side in Figure 6. As expected, they only agree for larger mR . To quantify this, we computed the relative error on

the approximation and plotted it logarithmically over a wider range of mR (Figure 7). The error decreases monotonically beyond $mR \gtrsim 5$, marking the onset of the semiclassical regime. (For $mR > 48$ the exact correlator becomes too small for stable numerics; a log-log plot extends the range.)

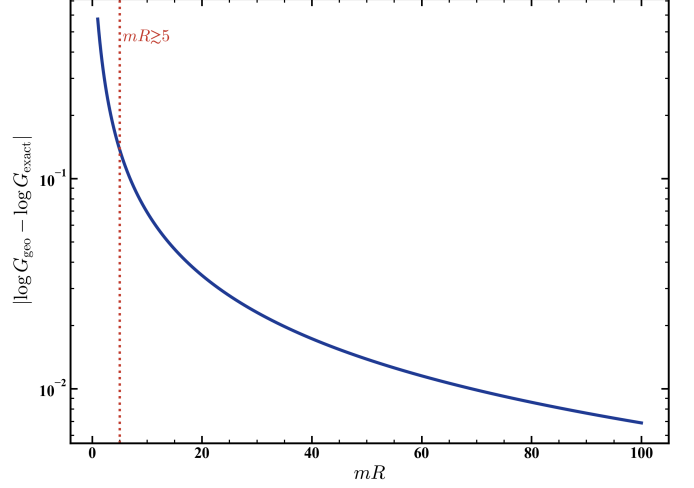


Figure 7: Logarithmic relative error of the geodesic approximation for varying mR , showing a clear drop for $mR \gtrsim 5$.

We also varied the boundary separation $\Delta x = |x_1 - x_2|$ at fixed mR (Figures 8 and 9). The error increases with separation (Figure 9), because for large Δx the geodesic penetrates deeper into the bulk where quantum fluctuations are more significant. The initial apparent re-convergence seen in the direct plot (Figure 8) is an artefact of both correlators becoming very small; the relative error (Figure 9) clearly shows the worsening.

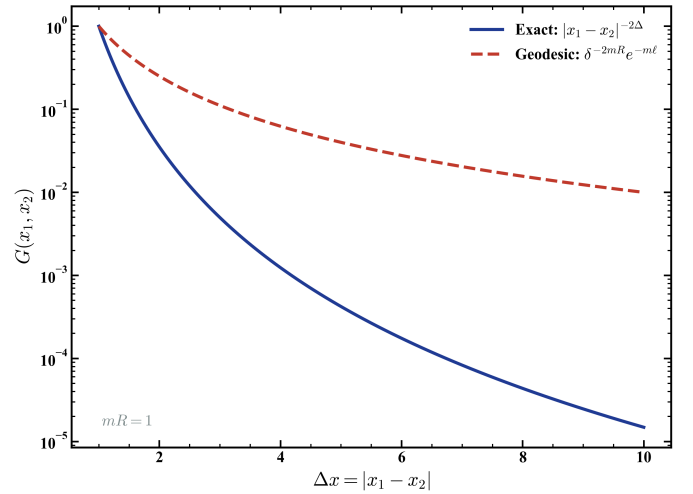


Figure 8: Exact and approximated correlators for varying separation Δx .

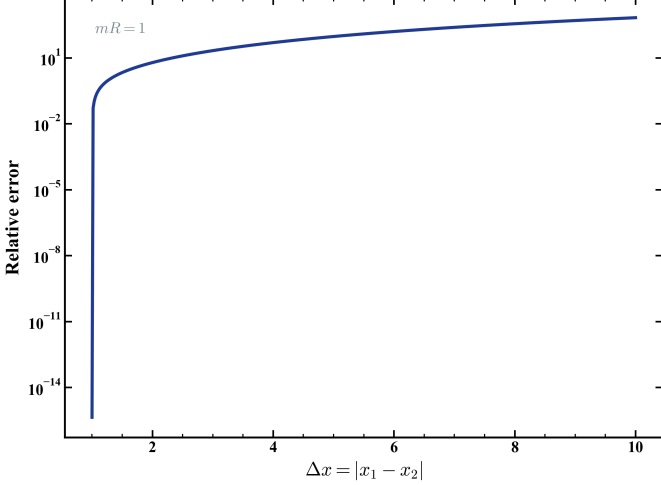


Figure 9: Relative error on the correlator approximation for varying Δx .

These numerical results confirm the geodesic approximation as a valid semiclassical description of CFT two-point correlators when bulk propagation is dominated by classical trajectories.

5.5 From continuum geodesics to the spectral heat kernel: saddle-point and exact identification

The numerical threshold $mR \gtrsim 5$ has a precise algebraic origin in the spectral holography (SH) framework.

SH two-point function. For a finite weighted graph $G = (V, E)$ with Laplacian L (§6.1), the two-point function between boundary vertices i, j is the massive resolvent

$$\langle \hat{O}_i \hat{O}_j \rangle = (L + \mu^2)_{ij}^{-1} = \int_0^\infty e^{-\mu^2 \sigma} K_\sigma(i, j) d\sigma \quad (47)$$

where $\mu = mR$ and $K_\sigma(i, j) = e^{-\sigma L}$ is the graph heat kernel.

Saddle-point limit. For graphs whose spectral embedding approximates a hyperbolic space, the heat kernel satisfies $K_\sigma(i, j) \sim A_\sigma e^{-d(i, j)^2/(4\sigma)}$ for $\sigma \ll d(i, j)^2$, with $d(i, j)$ the graph distance. Substituting into (47) and evaluating by steepest descent at $\sigma^* = d(i, j)/(2\mu)$ gives

$$(L + \mu^2)_{ij}^{-1} \stackrel{\mu d \gg 1}{\sim} C e^{-\mu d(i, j)}. \quad (48)$$

For a graph approximating AdS_3 , $d(i, j) \approx \ell(x_1, x_2)/R$, so $\mu d \gg 1$ is precisely $mR \gg 1$ – the semiclassical condition. The numerical onset $mR \gtrsim 5$ is where saddle-point dominance begins; for $mR < 5$ the full integral (47) is required.

Remark 5.1 (What the numerics probe). The numerical experiments compute the zero-temperature Poincaré-patch AdS_3 correlator $G \sim |x_1 - x_2|^{-2\Delta}$, which is *not* an SH saddle at any finite N : $D_{\text{bulk}} = N + 1$ requires $N = 2$ for a 3D bulk, giving the eternal BTZ black hole (P_2 graph), not Poincaré AdS_3 .

Remark 5.2 (Two distinct continuum limits). $D_{\text{bulk}} = N + 1$ holds at fixed finite N and counts the ambient dimension of $\mathcal{P} \subset \mathbb{R}^{N-1}$, the angular polytrope of section 6.3. The Poincaré-patch AdS_3 correlator probed numerically is not a finite- N SH saddle; it is the $N \rightarrow \infty$ Gromov–Hausdorff limit of hyperbolic graphs whose *intrinsic* Hausdorff dimension is 2 (the embedding dimension diverges; the intrinsic dimension stabilises). The two limits resolve cleanly: finite small N gives exact black-hole saddles ($P_2 \rightarrow \text{BTZ}$, $C_3 \rightarrow \text{SM}$, $K_{10} \rightarrow \text{M-theory}$); $N \rightarrow \infty$ recovers continuum AdS_d correlators via heat-kernel resummation (47). The numerical threshold $mR \gtrsim 5$ is the steepest-descent dominance point of (47) in the second regime; it is unrelated to the small- N saddles.

Conclusion of the numerical analysis. The numerics establish that the geodesic approximation becomes accurate for $mR \gtrsim 5$. This threshold marks the boundary between two analytically distinct regimes of the SH heat-kernel correlator (Remark 5.2); above it, the steepest-descent evaluation of $\int e^{-\mu^2 \sigma} K_\sigma(i, j) d\sigma$ is controlled, i.e., when $\mu d \gg 1$ or equivalently $mR \gg 1$. For $mR < 5$ the full integral is required. Thus the graph framework provides an exact, non-perturbative definition of the correlator that reduces to the continuum geodesic approximation in the large- mR limit, directly linking the numerical observation to the spectral data of the graph.

The SH framework is most exact at small N : BTZ ($N = 2$), the Standard Model ($N = 3$ with $\mathbb{Z}_2$ holonomy), and M-theory ($N = 10$ with trivial holonomy) all arise from finite graphs with $N \leq 10$. The $N \rightarrow \infty$ continuum limit serves as a consistency check, not the primary setting.

6 A Discrete Foundation for Holography

6.1 Modular structure, graph states, and the gap equation

Let $\mathcal{M}$ be a Type III₁ von Neumann factor [40] describing observables in a causal wedge [7]. The Tomita–Takesaki theorem [41] provides a modular automorphism group σ_t^ω for any faithful normal state ω , but for Type III₁ this group is *outer*, no self-adjoint $\hat{h} \in \mathcal{M}$ exists with $\sigma_t^\omega(a) = e^{it\hat{h}} a e^{-it\hat{h}}$. An observer confined to the wedge cannot construct a local clock.

The crossed product

$$\widehat{\mathcal{M}} = \mathcal{M} \rtimes_{\sigma^\omega} \mathbb{R} \quad (49)$$

adjoins the modular generator $\hat{h}_\omega = -\log \Delta_\omega$ as an affiliated self-adjoint operator. The crossed product is a Type II_∞ factor; it carries a canonical faithful semifinite trace Tr_{sf} and the modular flow becomes inner: $\sigma_t^\omega(a) = e^{it\hat{h}_\omega} a e^{-it\hat{h}_\omega}$. The semifinite trace is unique up to scale; fixing $\text{Tr}_{\text{sf}}(P) = M$ on a finite-trace projection P selects an observer reference frame in the sense of [42], with M the rank of the modular reference algebra. We identify this M with g_s^{-1} and the M-theory radius $R_{11} = M \ell_P$ in

§6.8; it is an observer scale, not a coupling constant. The KMS condition at $\beta = 2\pi$ identifies $\hat{h}_\omega$ with the Lorentz boost generator of the Rindler wedge at Unruh temperature $T = 1/(2\pi)$ [43–45]. The interpretation of the modular flow as a physical time direction realises, at the level of the crossed product, the thermal-time hypothesis of Connes and Rovelli [46].

6.1.1 Graph data and global state

Let $G = (V, E)$ be a finite connected graph with N vertices. Place a copy $\widehat{\mathcal{M}}_i$ of the crossed-product algebra at each vertex i . To each edge $\{i, j\} \in E$ assign:

- a *weight* $w_{ij} > 0$, interpreted as the von Neumann entropy of the bipartite state shared across the edge;
- an *automorphism* $\Phi_{ij} \in \text{Aut}(\mathfrak{g})$ with $\Phi_{ji} = \Phi_{ij}^{-1}$.

A global pure state $|\Psi_G\rangle$ on $\bigotimes_{i \in V} \mathcal{H}_i$ is constructed using a spanning tree $T \subset E$:

1. Choose a root $r \in V$ and orient every edge of T away from r .
2. Set $|\psi_r\rangle = |\Omega_0^{(r)}\rangle$, the vacuum of the root algebra.
3. Traverse T breadth-first. For each directed edge $i \rightarrow j$, apply a unitary U_{ij} on $\mathcal{H}_i \otimes \mathcal{H}_j$ that creates a bipartite state with fixed Schmidt spectrum $\{\lambda_\alpha^{(ij)}\}$, independent of the input state, with entropy $w_{ij} = -\sum_\alpha \lambda_\alpha^{(ij)} \log \lambda_\alpha^{(ij)}$.

The fixed-spectrum condition guarantees that $S(\rho_{i \cup j})$ is independent of the chosen spanning tree (Appendix A1 in the supplementary proofs).

Remark 6.1 (Spectral graphs versus tensor networks). Unlike a tensor network or its holographic incarnation (HaPPY codes [47], MERA [27]) - fixed graph, finite bond dimension, variational state [48] - each vertex here carries a Type II $_\infty$ factor with canonical trace and modular Hamiltonian. The edge weight w_{ij} is the bipartite von Neumann entropy, not a bond dimension, and the gap equation $w_{ij}^* = S(\rho_{i \cup j})$ enforces exact self-consistency. The bulk metric is derived algebraically from $L^\dagger$, not approximated. Edge automorphisms Φ_{ij} supply gauge holonomies; Theorem 6.5 counts massless gauge bosons as a topological invariant. Most importantly, the graph is summed over in the Spectral Ensemble (§6.8) - it is a dynamical variable, not a fixed background.

6.1.2 The admissible Laplacian cone (Stieltjes cone)

Define the *admissible Laplacian cone*

$$\mathcal{L}_+^G := \left\{ L \in \text{Sym}^{|V|} \left| L_{ij} \leq 0 \ (i \neq j), \sum_j L_{ij} = 0, L \succeq 0 \right. \right\} \quad (50)$$

with $L_{ij} = -w_{ij}$ for $\{i, j\} \in E$ and $L_{ii} = \sum_{j \sim i} w_{ij}$. For a connected graph, $\ker L = \text{span}\{\mathbf{1}\}$. The spectral decomposition is $L = \sum_{a=1}^{N-1} \lambda_a v^{(a)} v^{(a)T}$ with $0 <$

$\lambda_1 \leq \dots \leq \lambda_{N-1}$, and the Moore–Penrose pseudoinverse is $L^\dagger = \sum_{a=1}^{N-1} \lambda_a^{-1} v^{(a)} v^{(a)T}$.

Remark 6.2 (Why the Stieltjes cone). $\mathcal{L}_+^G$ is precisely the set of Laplacians whose pseudoinverse $L^\dagger$ is conditionally positive on $\mathbf{1}^\perp$, equivalent to L being the kernel of a Stieltjes integral $L_{ij} = \int_0^\infty (e^{-\sigma} - P_{ij}^\sigma) d\mu(\sigma)$ for a probability measure μ . This guarantees: (i) the angular Gram matrix $g_{ij} = (L^\dagger)_{ij} / \text{Tr } L^\dagger$ is positive semidefinite (Euclidean embedding exists); (ii) the heat semigroup $e^{-\sigma L}$ is stochastic (probabilistic interpretation of correlators); (iii) Schoenberg’s theorem applies [49], ensuring the spectral embedding lies in a Hilbert space. None of these hold for arbitrary symmetric matrices with non-positive off-diagonals; the cone constraint is the minimal algebraic structure for emergent geometry.

The dynamics of the graph is governed by the *spectral action* [18]

$$S_E = \text{Tr } f_E(D_G/\Lambda), \quad (51)$$

where D_G is the graph Dirac operator, f_E is a positive even function decaying rapidly at infinity (e.g., $f_E(x) = e^{-x^2}$), and Λ is a UV cutoff. The spectral action integrates out fermionic degrees of freedom and encodes both geometry and gauge fields. At the gap fixed point, the UV cutoff is fixed by the Planck self-gravity condition $\Lambda^2 = 2\lambda_1$ (see Theorem 6.7 and Appendix A6). This condition ties the cutoff to the spectral gap, rendering the theory UV-finite.

6.1.3 Gap equation and unique fixed point

The gap equation arises from modular equilibrium. The crossed-product KMS condition at $\beta = 2\pi$ (the Unruh temperature) requires that the modular Hamiltonian $\hat{h}_\omega$ generates the correct thermal correlations across every edge. This condition is satisfied if and only if the von Neumann entropy of the reduced state on the edge equals the weight w_{ij} that defines the entanglement spectrum. Thus consistency demands:

$$w_{ij}^* = S(\rho_{i \cup j}(\{w_{kl}^*\})) \quad \forall \{i, j\} \in E. \quad (52)$$

This is the *gap equation*: the condition that input weights equal the von Neumann entropies they generate.

Define the *gap map* $F : \mathbb{R}_{>0}^{|E|} \rightarrow \mathbb{R}_{>0}^{|E|}$ by $F_{ij}(\{w_{kl}\}) = S(\rho_{i \cup j}(\{w_{kl}\}))$. Using the Araki relative modular operator, the entropy can be expressed as a spectral integral over the modular spectral density $\rho_{ij}(\log \lambda; \{w_{kl}\})$:

$$F_{ij}(\{w_{kl}\}) = w_0 - \frac{g_{\text{YM}}^2}{2} \int_0^\infty \frac{d\lambda}{\lambda} \rho_{ij}(\log \lambda; \{w_{kl}\}) \text{Tr}[(d_\Phi \alpha)_{ij}^2], \quad (53)$$

where $\rho_{ij}(\log \lambda; \{w_{kl}\})$ is the spectral measure of $\log \Delta_{\omega_i | \omega_j}$ (the relative modular operator) and satisfies $\|\rho_{ij}\|_{L^1} \leq K$ uniformly in the weights. The twisted exterior derivative d_Φ is defined in §6.2.

Lemma 6.3 (Gap contraction and unique fixed point). *Let $\kappa = g_{\text{YM}}^2 K / (2\varepsilon)$ with $K = \|\rho_0\|_{L^1}$ and $\varepsilon > 0$ a lower bound on the weights. If $\kappa < 1$, then the gap map $F : [\varepsilon, w_0]^{|E|} \rightarrow [\varepsilon, w_0]^{|E|}$ is a strict contraction*

with $\|DF\|_{\ell^\infty \rightarrow \ell^\infty} \leq \kappa$, and has a unique fixed point $\{w_{ij}^*\} \subset \mathcal{L}_+^G$ depending smoothly on g_{YM} .

The hypothesis $\kappa < 1$ is not needed for:

- (i) trivial holonomy ($\kappa = 0$ identically, e.g. K_{10});
- (ii) discrete holonomy (e.g. $\mathbb{Z}_2$ on C_3);
- (iii) $g_{\text{YM}} = 0$ ($F \equiv w_0$).

Proof. See Appendix A2. $\square$

6.1.4 Quantum Markov Property and global time

For a tripartition $A, B, C \subset V$ where B separates A from C (every path from A to C passes through B), define the conditional mutual information

$$I(A : C \mid B) = S(AB) + S(BC) - S(B) - S(ABC), \quad (54)$$

where $S(X)$ denotes the von Neumann entropy of the reduced state on vertices X .

Theorem 6.4 (Exact QMP on trees). *If G is a tree ($\beta_1 = 0$), then $I(A : C \mid B) = 0$ for every separating tripartition.*

Proof. Supplementary proof Appendix A3 (induction on leaves using the fixed-Schmidt condition). $\square$

Conjecture 6.1 (QMP for cycles). *Let G be a connected graph at its gap fixed point with $\beta_1(G) \geq 1$ and $\kappa < 1$. For every tripartition $A, B, C \subset V$ with B separating A from C ,*

$$I(A : C \mid B)_{\omega_G} \leq C(G) e^{-2\pi\lambda_1/g_{\text{YM}}^2}, \quad (55)$$

with $C(G)$ depending only on the graph topology, not on the weights or holonomies at the fixed point. The conjecture is a theorem when $\beta_1=0$ or $g_{\text{YM}}=0$ (trivial cocycle). The general case reduces to bounding the holonomy contribution to the cocycle $(D\omega_i; D\omega_j)_t$ around independent cycles by the modular spectral norm; this bound is established in toy examples (single cycles with $\mathbb{Z}_n$ holonomy) but the general case is open.

The exponential form is forced by KMS analyticity: the cocycle extends to a holomorphic function on the $\beta = 2\pi$ strip, and the cycle-holonomy obstruction inherits this width. Trivialisation on every edge (Appendix A4) leaves only the cycle holonomy as obstruction, hence the $e^{-2\pi\lambda_1/g_{\text{YM}}^2}$ suppression.

Modular equilibrium at the gap fixed point implies $\omega_{ij}(\log \Delta_{\omega_i|\omega_j}) = 0$ for every edge, which is equivalent to the cocycle $(D\omega_i : D\omega_j)_t$ being the identity for all t . Consequently, the local modular flows are perfectly synchronised across edges. Patching these local flows together using the QMP yields a unique global timelike Killing vector field on the emergent bulk (Thm. 6.7). Thus the crossed-product modular clocks, the gap fixed point, and the QMP jointly produce a classical background space-time from purely algebraic data. For graphs with discrete holonomy (e.g. $\mathbb{Z}_2$ on C_3), $\kappa=0$ identically and the bound is trivial; the conjecture has content only for graphs with continuous cycle holonomies (e.g. the $\beta_1=36$ moduli of K_{10} under non-trivial flux).

6.2 Gauge from holonomy: the holonomy kernel theorem

The edge automorphisms $\Phi_{ij} \in \text{Aut}(\mathfrak{g})$ compose around cycles to generate the *holonomy group* $\mathcal{H} \subseteq \text{Aut}(\mathfrak{g})$. Fix a representation (ρ, V_ρ) of $\text{Aut}(\mathfrak{g})$. The *twisted graph Laplacian* acts on $\alpha \in V_\rho^N$ as

$$(\Delta_{\Phi, \rho} \alpha)_i = \sum_{j \sim i} w_{ij} (\alpha_i - \rho(\Phi_{ij}) \alpha_j). \quad (56)$$

A *parallel section* satisfies $\alpha_i = \rho(\Phi_{ij}) \alpha_j$ for every edge $\{i, j\}$; these are the discrete analogues of massless gauge fields.

Theorem 6.5 (Holonomy kernel theorem). *For any connected graph and any representation (ρ, V_ρ) ,*

$$\ker \Delta_{\Phi, \rho} \cong \text{Fix}(\mathcal{H}|_\rho) := \{v \in V_\rho : \rho(P)v = v \ \forall P \in \mathcal{H}\}. \quad (57)$$

In particular, $\dim \ker \Delta_{\Phi, \rho} = \dim \text{Fix}(\mathcal{H}|_\rho)$ is independent of all edge weights $\{w_{ij}\}$.

Proof. Fix a spanning tree T and root v_0 . Every parallel section is uniquely determined by $\alpha_{v_0} = v$ via $\alpha_i = \rho(P_{v_0 \rightarrow i}^T)v$, where $P_{v_0 \rightarrow i}^T$ is the tree-path holonomy. For each non-tree edge, the cycle condition imposes $\rho(P_\gamma)v = v$ for the cycle holonomy $P_\gamma \in \mathcal{H}$. Hence parallel sections biject with $\text{Fix}(\mathcal{H}|_\rho)$. The weights w_{ij} appear only in $\langle \alpha, \Delta_{\Phi, \rho} \alpha \rangle = \sum_{\{i, j\}} w_{ij} \|\alpha_i - \rho(\Phi_{ij}) \alpha_j\|^2$, which scales non-zero eigenvalues but cannot alter $\ker \Delta_{\Phi, \rho}$. (Full detail in Appendix A5.) $\square$

6.2.1 Example: Standard Model gauge group from the triangle graph

Take C_3 with $\mathfrak{g} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)_Y$. Assign $\Phi_{12} = \Phi_{23} = \text{Id}$ and $\Phi_{31} = \tau$, the $\mathbb{Z}_2$ outer automorphism that exchanges the $\mathfrak{su}(2)$ and $\mathfrak{u}(1)_Y$ factors while fixing $\mathfrak{su}(3)$; its fixed subalgebra is $\text{Fix}(\tau) = \mathfrak{su}(3) \oplus \mathfrak{u}(1)_Q$, the algebra of the Standard Model's strong and electromagnetic interactions. The unique cycle has holonomy $\mathcal{H} = \langle \tau \rangle \cong \mathbb{Z}_2$. Theorem 6.5 then gives

$$\dim \ker \Delta_{\Phi, \rho} = \dim \text{Fix}(\tau) = 8 + 1 = 9, \quad (58)$$

independent of all edge weights. The nine massless states are the eight gluons and the photon - the Standard Model massless gauge sector, with no fine-tuning required.

6.2.2 Discrete Higgs mechanism

In the continuum electroweak theory, the $W^\pm$ and Z bosons acquire masses by “eating” Goldstone modes of a scalar field that breaks $SU(2) \times U(1) \rightarrow U(1)_Q$. In the graph framework, the same mass generation occurs without any scalar field. The $\mathbb{Z}_2$ holonomy τ on C_3 projects out the $W^\pm$ and Z components from the kernel of the twisted Laplacian $\Delta_{\Phi, \rho}$. The remaining generators of $\mathfrak{g}$ are not fixed by τ ; they become massive, with mass scale set by the spectral gap λ_1 . For homogeneous C_3 with $w_{ij} = w$, the non-zero eigenvalues on the $\mathfrak{su}(2)$

part are of order w , and $w = w^*$ is determined by the gap equation. The role of the Higgs vacuum expectation value is played by $v = \sqrt{\lambda_1}$ (dimension of mass; recall $[\lambda_1] = [\text{mass}]^2$ from $G_N = 1/(4\lambda_1)$). At the homogeneous saddle, the odd-sector twisted Laplacian eigenvalues $\{w_0, w_0, 4w_0\}$ give $m_W^2, m_Z^2 \propto \lambda_1$ with ratio $m_Z/m_W = 2$ when $w_{31}^* = w_{12}^*$.

The non-trivial flux $\|d_\Phi \alpha\|_{31}^2 = \|\tau - \text{Id}\|^2$ in the gap-map source term (53) is expected to drive $w_{31}^*/w_{12}^* \rightarrow \cos^2 \theta_W$ to leading order in κ , yielding $m_W/m_Z = \cos \theta_W$ as a *derived* relation (the Standard Model tree-level prediction). This is a [P]-tier result: the linearised gap equation around the gauged C_3/σ' saddle must be solved explicitly to confirm the prefactor; the calculation is nontrivial and left for future work. The full derivation of $\cos \theta_W$ from the gap equation would tie the electroweak mixing angle to the ratio of graph edge weights at the fixed point, providing a purely combinatorial origin for the Weinberg angle.

6.3 Gravity, symmetry branes, and perturbations

The Moore–Penrose pseudoinverse $L^\dagger$ of the graph Laplacian encodes the emergent geometry. Define the *angular metric*

$$g_{ij} = \frac{(L^\dagger)_{ij}}{\text{Tr}(L^\dagger)}, \quad (59)$$

which is scale-invariant under uniform rescaling of edge weights. The *spectral embedding*

$$\mathbf{x}_i = (\lambda_a^{-1/2} v_i^{(a)})_{a=1}^{N-1} \in \mathbb{R}^{N-1} \quad (60)$$

(where $v^{(a)}$ are orthonormal eigenvectors of L with eigenvalues λ_a) maps each vertex to a point. The convex hull $\mathcal{P} = \text{conv}\{\mathbf{x}_i\}$ is a piecewise-flat polytope: the Riemann tensor vanishes on the interior of each simplex face, while distributional curvature (conical singularities) concentrates at vertices and lower-dimensional strata. The angular polytope $\mathcal{P}$ supplies the $(N-1)$ angular dimensions of the bulk; the time and radial directions arise separately, from the modular and spectral structure of the vertex algebras respectively.

Time from modular flow. At each vertex, the crossed-product construction (§6.1) makes the Tomita–Takesaki modular flow σ_t^ω inner, generated by $\hat{h}_\omega = -\log \Delta_\omega$. At the gap fixed point the cocycle trivialises on every edge (Appendix A4), synchronising local modular clocks; under the QMP these patch to a global Killing time t . This is the algebraic realisation of the Connes–Rovelli thermal-time hypothesis [46].

Radial direction from spectral RG flow. The radial coordinate r emerges in the standard holographic RG sense: the boundary regulator $\delta \rightarrow 0$ at $r \rightarrow \infty$ plays the UV role, and the bulk interior $r \rightarrow 0$ the IR role. Concretely, r is built from the heat-kernel parameter σ of the resolvent $(L + m^2)^{-1} = \int e^{-m^2 \sigma} e^{-\sigma L} d\sigma$ (§5.5): short σ resolves the UV graph structure, long σ probes the IR spectral gap, mirroring the holographic RG scaling $r \sim \sigma^{-1/2}$ familiar from AdS/CFT.

Remark 6.6 (Extremal vertices are timelike infinities). A vertex i is extremal in $\mathcal{P}$ iff $\mathbf{x}_i$ lies on its convex hull, which holds iff the modular Hamiltonian $\hat{h}_i$ has a one-sided spectrum (the boundary condition for a wedge algebra). Under the spectral embedding, extremal vertices map to the timelike infinities $i^\pm$ of the emergent bulk: the asymptotic data of each boundary CFT is localised there. The number N_∂ of extremal vertices is therefore precisely the conformal invariant N of Remark 2.1, and the stratum invariant $\alpha = (N_\partial + 1 + n_{\text{mod}} - n_{\text{gh}})/2$ of §6.3.6 depends on it directly.

The distributional scalar curvature of $\mathcal{P}$ is a sum of delta-function contributions at the singular loci, with coefficients given by the angular deficits $2\pi - \sum_F \theta_F(p)$, where $\theta_F(p)$ are the interior dihedral angles of the faces meeting at p . We write $\hat{R}_\mathcal{P}$ for this distributional curvature. The locus $\ker \hat{R}_\mathcal{P} \subseteq V_\rho$, the subspace on which the conical-singularity sources vanish, plays a central role later: by the graph Weitzenböck identity (Thm. 6.11), it controls the supersymmetry count via $\ker D_G \cong \text{Fix}(\mathcal{H}|_\rho) \cap \ker \hat{R}_\mathcal{P}$. A regular $(N-1)$ -simplex (e.g. $\mathcal{P}_{K_{10}} = \Delta^9$) has no interior conical singularities, so $\hat{R}_\mathcal{P} \equiv 0$ and full SUSY is preserved; a polytope with conical defects (e.g. $\mathcal{P}_{C_3}$ with its $\mathbb{Z}_3$ centroid singularity) reduces $\ker D_G$ accordingly.

The full bulk spacetime is therefore a warped product of r , t , and $\mathcal{P}$, with bulk spatial dimension

$$D_{\text{bulk}} = 2 + (N - 1) + \dim \mathcal{M}_{\text{int}}, \quad (61)$$

where the $+2$ counts the radial direction and time, $(N - 1) = \dim \mathcal{P}$ counts the angular dimensions, and $\dim \mathcal{M}_{\text{int}}$ counts internal dimensions coming from KK towers generated by cycle holonomies (each independent cycle contributes a compact direction with radius set by the edge weights).

Assuming the existence of a global timelike Killing vector (derived from modular equilibrium and the Quantum Markov Property, see Appendix A13), the emergent bulk metric takes the warped product form

$$ds^2 = -f(r) dt^2 + f(r)^{-1} dr^2 + r^2 g_{ij} d\theta^i d\theta^j + ds_{\text{int}}^2, \quad (62)$$

where g_{ij} is the angular metric above, r is the radial coordinate above, $f(r)$ is a function determined by the angular polytope's curvature, and ds_{int}^2 describes the internal KK dimensions. The precise form of $f(r)$ follows from the vacuum Einstein equations, which are derived thermodynamically in Appendix A13.

Theorem 6.7 (Gravitational constants [C]). *Under the QMP and the wedge-algebra assumption, at the gap fixed point of any connected graph with $N \geq 2$ extremal vertices,*

$$G_N = \frac{1}{4\lambda_1}, \quad \Lambda_G = -\frac{N\lambda_1}{2(N-1)}, \quad \ell_G^2 = \frac{(N-1)^2}{\lambda_1}. \quad (63)$$

Both constants are derived from the heat-kernel expansion $S_E \supset \frac{f_2 \Lambda_{\text{UV}}^2}{8\pi^2} \int_{\mathcal{P}} R$ matched to Einstein–Hilbert at the self-gravity cutoff $\Lambda_{\text{UV}}^2 = 2\lambda_1$ with $f_2 = (2\pi)^{-1}$ and $\int_{\mathcal{P}} R d\text{vol} = 4\pi \chi(\mathcal{P}) = 4\pi$. The cosmological constant follows from $\Lambda_G = -N(N-1)/(2\ell_G^2)$ in $D = N+1$ dimensions.

The derivation matches the spectral action $\text{Tr } f_E(D_G/\Lambda)$ to the Einstein–Hilbert action via the Connes–Chamseddine heat-kernel expansion, using the self-gravity UV cutoff $\Lambda^2 = 2\lambda_1$ and the Gauss–Bonnet evaluation $\int_{\mathcal{P}} R \text{dvol} = 4\pi$ (since $\chi(\mathcal{P}) = 1$). A detailed proof is given in Appendix A6.

6.3.1 Spectral gap field and mass parameter

The global spectral gap λ_1 alone does not capture spatial inhomogeneities. Define the *local spectral gap field* $\Phi_G(i)$ as the smallest non-zero eigenvalue of the Laplacian restricted to the closed star of vertex i , extended piecewise-linearly to $\mathcal{P}$. Its deviation $\delta\Phi(i) = \Phi_G(i) - \lambda_1$ satisfies $\sum_i \delta\Phi(i) = 0$. The mass parameter μ appearing in the warp factor

$$f(r) = \frac{r^2}{\ell_G^2} - \frac{\mu}{r^{N-2}} \quad (64)$$

is given by

$$\mu = \frac{(N-1)^2}{\lambda_1} \sum_{i \in V} \|\nabla_{\mathcal{P}} \Phi_G(x_i)\|^2. \quad (65)$$

When Φ_G is constant ($\mu = 0$) the geometry is pure AdS; $\mu > 0$ signals a localised “hot spot” that sources a black hole.

6.3.2 Example diagrams

Two principal small- N saddles illustrate the warped-product construction: the path graph P_2 produces the BTZ wormhole (Fig. 10), and the triangle C_3 produces a Y-shaped trinion with a $\mathbb{Z}_3$ conical brane at the centroid (Fig. 11).

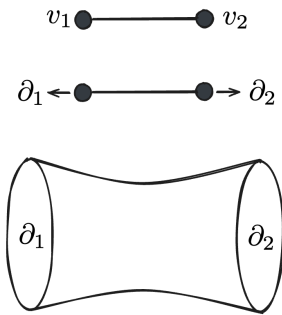


Figure 10: BTZ from P_2 : (D1) two vertices, one edge $w^* = S_{\text{BH}}$; (D2) angular polytope = line segment, boundaries ∂_1, ∂_2 ; (D3) inflated cylinder = eternal BTZ wormhole. No cycles $\rightarrow$ exact Markovianity. ¹

¹Figures created with <https://excalidraw.com/> and Adobe Photoshop.

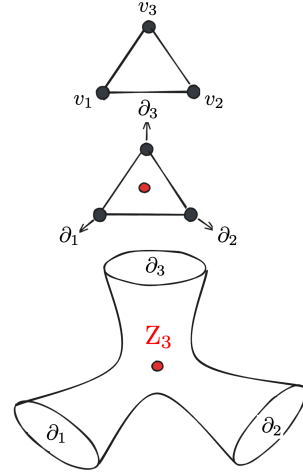


Figure 11: Triangle C_3 : (D1) equilateral triangle, three vertices; (D2) equilateral polytope, $\mathbb{Z}_3$ conical brane at centroid (red dot); (D3) Y-shaped junction (trinion). Cycle $\beta_1 = 1$ breaks global time.

6.3.3 Symmetry branes

Two classes of brane arise naturally from automorphisms of G and appear as singular loci in the emergent geometry.

Domain walls. An admissible reflection $\sigma \in \text{Aut}(G)$ ($\sigma^2 = \text{Id}$, $\det \sigma = -1$, preserving weights and holonomies) defines a codimension-one wall at the σ -fixed locus. The wall is stable if weight perturbations preserve σ -symmetry; the antisymmetric component δw^- dissolves the wall discontinuously through obstruction to the conditional expectation $\mathcal{M}_G \rightarrow \mathcal{M}_G^+$.

Conical branes. A $\mathbb{Z}_k$ rotation fixing an interior point $p_0 \in \mathcal{P}$ produces a codimension-two conical singularity with deficit angle $\delta = 2\pi(1 - 1/k)$ and tension $T_0 = (k-1)\lambda_1/k$, depending only on k and λ_1 . The brane persists under $\mathbb{Z}_k$ -equivariant perturbations and dissolves only when the symmetry is broken (the curvature smooths and the singularity vanishes).

Full stability analysis is given in Appendix A10. The $\mathbb{Z}_3$ centroid singularity of C_3 (Fig. 11) and the ghost vertex of the gauged C_3/σ' saddle (Rmk. 6.9) are both instances of this construction.

6.3.4 Islands from spectral-gap minimisation

When the local spectral gap $\Phi_G(i)$ drops below the threshold λ_1^{thr} at which $[\mathbf{D}, \hat{h}_i] \neq 0$, vertex i becomes a de Sitter bubble and is eligible as an island. Define

$$V_{\text{isl}} := \{i \in V \setminus V_{\partial} : \Phi_G(i) < \lambda_1^{\text{thr}}\}. \quad (66)$$

A *cut* is a set of edges whose removal disconnects the graph; its total weight is the sum of edge weights. For a set of vertices X , $\gamma_X^{\min}$ denotes the cut of minimum total

weight that separates X from its complement (the minimal cut).

Proposition 6.8 (Discrete island formula). *For any boundary region $A \subseteq V_{\text{isl}}$,*

$$S(A) = \min_{I \subseteq V_{\text{isl}}} \frac{1}{4G_N} \sum_{\{i,j\} \in \gamma_{A \cup I}^{\min}} \bar{\Phi}_G(i,j), \quad (67)$$

where $\gamma_{A \cup I}^{\min}$ is the minimal cut separating $A \cup I$ from its complement. When $V_{\text{isl}} = \emptyset$, $I = \emptyset$ recovers the RT formula.

Sketch. Subadditivity gives $S(A) \leq S(A \cup I)$; applying RT to $A \cup I$ yields the upper bound. Any $i_0 \notin V_{\text{isl}}$ has $w_{i_0 j}^* \geq (8G_N)^{-1} \lambda_1^{\text{thr}}$, so encircling it increases the cut weight. Hence the optimal $I^* \subseteq V_{\text{isl}}$, and the bound is saturated. The full argument is in Appendix A8. $\square$

As the mass parameter $\mu(t)$ grows, vertices enter V_{isl} sequentially; the minimal cut jumps when wrapping a new bubble becomes cheaper than cutting through it – the discrete quantum extremal surface transition that produces the Page curve. For a gauged admissible reflection σ , the ghost vertex at $F = \text{Fix}(\sigma)$ can anchor cuts, adding a wall-tension term $T_{\text{wall}} \ell(\gamma_{A \cup I}^{\min} \cap F)$ to the minimisation, with $T_{\text{wall}} = \kappa_F \lambda_1 / (2\pi)$.

Remark 6.9 (Ghost-vertex algebra type). When an admissible reflection $\sigma \in \text{Aut}(G)$ is gauged, the quotient G/σ carries a codimension-two ghost vertex at the fixed-point locus $F = \text{Fix}(\sigma)$. The local algebra at this vertex is not Type II_∞ but Type II_1 [42]: the trace Tr_F inherits the finite normalisation from σ -invariance, with $\dim_{\text{II}_1} \mathcal{M}_F = e^{S_{\text{as}}}$ [42, 50]. The local spectral gap is $\lambda_1^{\text{eff}}(p_{\text{gh}}) = \lambda_1^{\text{cover}}/2 + O(M^{-1})$, automatically below the de Sitter threshold λ_1^{thr} that triggers island formation (§6.3.4). This places the ghost vertex permanently in V_{isl} without tuning, making it the natural locus for cosmological-constant suppression.

6.3.5 Metric-mode rigidity at finite N

A physical metric mode is a trace-constrained weight perturbation δw_{ij} ($\sum \delta w_{ij} = 0$) producing a non-trivial change in the warp factor $f(r, \theta)$. Scalar modes shift the mass parameter, $\delta f(r) = \delta \mu \cdot r^{1-n}$; angular modes carry non-trivial dependence on the polytope eigenfunctions, $\delta f(r, \theta) = \sum_{a \geq 1} \alpha_a(r) Y_a(\theta)$ with $-\Delta_{\mathcal{P}} Y_a = \mu_a Y_a$, $\mu_a > 0$.

Theorem 6.10 (Metric-mode rigidity). *At any finite N , every trace-constrained weight perturbation of the gap fixed point is a scalar mode: $\#\{\text{physical metric modes}\} = 1$. In the densification limit $N_k \rightarrow \infty$ with $\mathcal{P}_k \rightarrow (\mathcal{M}_\infty, \gamma^{(0)})$ of dimension $d = \lim(N_k - 1)$, Weyl’s law gives $\#\{\text{angular modes below } \Lambda\} \sim \omega_{d-1}(2\pi)^{-(d-1)} \text{Vol}(\mathcal{M}_\infty) \Lambda^{d-1} \rightarrow \infty$.*

Sketch. The linearised warp-factor equation for a mode of eigenvalue μ_a is $\alpha_a'' + (n/r)\alpha_a' - (\mu_a/r^2)\alpha_a = S_a(r)$. On a finite graph $\mathcal{P}$ is piecewise flat, so the linearised Einstein

tensor vanishes on each simplex interior, forcing $S_a \equiv 0$ for $a \geq 1$; with $\alpha_a(\infty) = 0$ the only solution is $\alpha_a = 0$. The continuum limit restores non-zero sources for every Laplace–Beltrami eigenfunction, and Weyl’s law gives the stated counting. A complete verification using the Lichnerowicz formula on piecewise-flat polytopes is given in Appendix A7. $\square$

At finite N , weight perturbations therefore reduce to a global mass shift, completely captured by the perturbed gap equation. This rigidity is the dynamical counterpart of the algebraic contraction of the conformal group (§3.3): K_a has no angular modes to act on.

6.3.6 One-loop correction: the stratum invariant α

Theorem 6.7 gave the classical constants (G_N, Λ_G) from the leading heat-kernel coefficient. The next-order correction comes from quadratic fluctuations around the gap saddle, and its sign on the partition function fixes the logarithmic entropy correction. The relevant fluctuation modes are dictated by the SSCA: weight perturbations on $\mathcal{L}_+^G$ (constrained to one scalar mode at finite N by Thm. 6.10, plus N_∂ asymptotic charges); $U(1)$ holonomy phases on each independent cycle (one per generator of $H_1(G, \mathbb{Z})$); and Faddeev–Popov ghosts [51, 52] gauge-fixing the residual boundary diffeomorphisms.

Each bosonic Gaussian contributes $\lambda_1^{-1/2}$ (the Hessian scales linearly in λ_1 by the same dimensional argument that gave $G_N \propto \lambda_1^{-1}$ in Thm. 6.7); each anticommuting ghost contributes $\lambda_1^{+1/2}$. Combining:

$$Z_{1\text{-loop}} \propto \lambda_1^{-\alpha/2}, \quad \alpha = \frac{1}{2}(N_\partial + 1 + n_{\text{mod}} - n_{\text{gh}}), \quad (68)$$

where $N_\partial + 1$ counts the trace-constrained asymptotic charges plus the global mass shift, $n_{\text{mod}} = \beta_1(G)$ counts continuous cycle moduli, and $n_{\text{gh}} = N_\partial - 1$ for continuous holonomy and 0 for discrete. Since $S_{\text{BH}} = \mathcal{A}/(4G_N) \propto \lambda_1^{-1}$ (Thm. 6.7), this translates directly into

$$\Delta S = \alpha \log S_{\text{BH}} + O(1). \quad (69)$$

All three counts are topological invariants of the graph stratum, so α is independent of λ_1 , of the trace rank M , and of detailed saddle data, a sharp UV-complete observable.

G	N_∂	n_{mod}	n_{gh}	α	status
P_2 (BTZ)	2	1	1	3/2	calibrates [52]
$C_3/\mathbb{Z}_2$	2	0	0	3/2	EW vacuum
C_3	3	0	0	2	new (3-boundary)
K_{10}	10	36	9	19	new (M-theory)

The P_2 entry reproduces Sen’s BTZ log-correction [52], calibrating the framework. The coincidence $\alpha_{C_3/\mathbb{Z}_2} = \alpha_{P_2} = 3/2$ is structural: gauging the admissible reflection σ' on C_3 reduces it to P_2 in modular data, and α depends only on $(N_\partial, n_{\text{mod}}, n_{\text{gh}})$. The ungauged C_3 saddle

$\alpha=2$ is therefore the genuinely new three-boundary prediction. For K_{10} , the $\beta_1=36$ cycle moduli give $\alpha=19$; this correction is invisible to BFSS at any finite order in $g_s=M^{-1}$ because it arises from trans-stratal measure (Rmk. 6.15), not from fluctuations within the $N=10$ stratum. Derivation in Appendix A9.

6.4 The graph Dirac operator and the SSCA

Define the *graph Dirac operator*

$$(D_G\psi)_i = \sum_{j \sim i} \sqrt{w_{ij}} \gamma_{ij} \rho(\Phi_{ij}) \psi_j, \quad \gamma_{ij} = \sum_a (x_i^a - x_j^a) \gamma^a, \quad (70)$$

where γ^a generate the Clifford algebra $\text{Cl}(N-1)$ with $\{\gamma^a, \gamma^b\} = 2\delta^{ab}$. Identify $Q = D_G$ as the supercharge.

Place Grassmann-valued spinor fields $\psi_i, \bar{\psi}_i$ at each vertex, transforming under the spinor-gauge bundle $\mathcal{S} \otimes V_\rho$. The fermionic action is the quadratic form

$$S_{\text{Dirac}}[\bar{\psi}, \psi] = \sum_{i \in V} w_i \bar{\psi}_i (D_G\psi)_i, \quad (71)$$

where $w_i = \sum_{j \sim i} w_{ij}$. Integrating out the fermions gives the Pfaffian $\text{Pf}(D_G) = \exp(\frac{1}{2} \text{Tr} \log D_G)$, which by a Mellin transform yields the spectral action [18]:

$$\begin{aligned} \text{Tr} \log D_G &= - \int_0^\infty \frac{d\sigma}{\sigma} \text{Tr} e^{-\sigma D_G^2} \\ &\Rightarrow \det D_G = e^{-\text{Tr} f_E(D_G/\Lambda)}. \end{aligned}$$

The spectral action thus encodes the quantum dynamics of the fermionic matter and provides the measure for the spectral ensemble.

Theorem 6.11 (Graph Weitzenböck identity). *At any gap fixed point,*

$$Q^2 = D_G^2 = \Delta_{\Phi, \rho} + \frac{1}{4} \hat{R}_{\mathcal{P}} + F_\Phi, \quad (72)$$

where $\hat{R}_{\mathcal{P}}$ is the distributional scalar curvature of $\mathcal{P}$ (supported on conical singularities), and $F_\Phi = \sum_\gamma \rho(P_\gamma - \text{Id})$ is the holonomy flux. All three terms are positive semidefinite, so

$$\ker Q = \ker \Delta_{\Phi, \rho} \cap \ker \hat{R}_{\mathcal{P}} \cong \text{Fix}(\mathcal{H}|_\rho) \cap \ker \hat{R}_{\mathcal{P}}, \quad (73)$$

giving the exact SUSY count $\mathcal{N}_{\text{SUSY}} = \dim \ker Q$.

Proof. Write $D_G = \sum_a \gamma^a \nabla_a$ in a local orthonormal frame of the spectral embedding. Squaring gives the grade-0 term $\sum_a \nabla_a^2 = \Delta_{\Phi, \rho}$ and grade-2 terms from $[\nabla_a, \nabla_b]$. The latter split: curvature supported on singular strata of $\mathcal{P}$ gives $\frac{1}{4} \hat{R}_{\mathcal{P}}$; curvature supported on non-contractible cycles gives F_Φ . Orthogonality follows from different Clifford grades and disjoint support. Full detail in Appendix A11. $\square$

For C_3 with $\mathbb{Z}_2$ holonomy, $\mathcal{N}_{\text{SUSY}} = 0$ (no parallel sections fixed by τ survive the curvature condition). For K_{10} with trivial holonomy, $\text{Cl}(9) \cong \text{Mat}_{16}(\mathbb{R}) \oplus \text{Mat}_{16}(\mathbb{R})$ and $\mathcal{N}_{\text{SUSY}} = 16$, the maximal supersymmetry of M-theory.

6.4.1 Momentum and curvature operators

Let X^a be the diagonal matrix of spectral embedding coordinates: $(X^a)_{ij} = x_i^a \delta_{ij}$, acting on the vertex Hilbert space $\mathbb{C}^N$. Define the momentum operators

$$P_a = -i[L, X^a]. \quad (74)$$

A direct computation using the commutator $[L, X^a]$ yields matrix elements $(P_a)_{ij} = -i(x_i^a - x_j^a) L_{ij}$. The commutator of two momenta gives the emergent Riemann curvature:

$$[P_a, P_b] = \mathcal{R}_{ab}, \quad (75)$$

where $\mathcal{R}_{ab}$ is the graph curvature operator, whose matrix elements are supported on the conical singularities of $\mathcal{P}$. A derivation is provided in Appendix A12.

6.4.2 The spectral-superconformal algebra (SSCA)

The Poincaré-dilatation algebra anticipated in §3.3 closes on the graph with the addition of the supercharge $Q = D_G$ and the internal gauge generators T^A . The (anti)closure relations are

$$[P_a, D_G] = \Gamma_a \mathcal{G}_a, \quad (76)$$

$$[T^A, D_G] = \sum_{\{i,j\}} \sqrt{w_{ij}^*} \Gamma_{ij} \otimes \rho([T^A, \Phi_{ij}]), \quad (77)$$

$$\{D_G, D_G^\dagger\} = 2\Delta_{\Phi, \rho} + 2\mathcal{R}_{\mathcal{P}}, \quad (78)$$

together with $[\mathbf{D}, D_G] = \frac{1}{2} D_G$ for the heat-kernel dilatation $\mathbf{D} = \sigma d/d\sigma$. The three brackets carry distinct physical content: $[P_a, D_G] \neq 0$ is gravity (translations curve under the supercharge because the spectral embedding is curved); $[T^A, D_G] \neq 0$ is gauge interaction (holonomies rotate gauge fields); $\ker D_G$ gives supersymmetry, with $\mathcal{N}_{\text{SUSY}} = \dim \ker D_G$ via Thm. 6.11. The triple $(\mathbf{D}, D_G, D_G^\dagger)$ closes on a discrete $\mathfrak{osp}(1|2)$ subalgebra, with $\{D_G, D_G^\dagger\} = 2H_{\text{grav}}$ the bosonic generator - the supersymmetric completion of the Poincaré-dilatation algebra, exact at every finite N .

The conformal supercharge S_α of weight $-\frac{1}{2}$ exists at finite N only as the formal $\mathbf{D}$ -dual of D_G ; the inversion mapping $Q \rightarrow S$ is forbidden by the same boundedness that forbids K_a (§3.3). The SSCA is therefore a genuine algebraic contraction of $\mathfrak{osp}(\mathcal{N}|4)$ at finite N , with the full superconformal algebra recovered only as $N \rightarrow \infty$; the K_{10} saddle's continuum limit must reproduce $\mathfrak{osp}(8|4)$ as a consistency check (§6.8).

At the gap fixed point the Connes-Radon-Nikodym cocycle trivialises on every edge (Prop. A4), so $[\mathbf{D}, \hat{h}_i] = 0$ for all extremal vertices and the local one-sided modular clocks synchronise into a global time direction. When $\Phi_G(i)$ drops below threshold, $[\mathbf{D}, \hat{h}_i] \neq 0$, synchronisation breaks, and a de Sitter bubble nucleates (§6.3.4).

6.4.3 Modular conjugation and CPT

The Tomita operator $S_i = J_i \Delta_i^{1/2}$ gives an antiunitary modular conjugation satisfying $J_i^2 = \epsilon_i \mathbf{1}$, $J_i \hat{h}_i J_i = -\hat{h}_i$,

$J_i \mathcal{M}_i J_i = \mathcal{M}'_i$. The reversal of modular time is the algebraic CPT operation. At the gap fixed point the cocycle trivialises on every edge (Prop. A4), so $J_i|_{\mathcal{M}_{ij}} = J_j|_{\mathcal{M}_{ij}}$; by connectedness these patch to a global J_G with $J_G D_G J_G^* = \epsilon' D_G$. The signs (ϵ, ϵ') fix the KO-dimension of the spectral triple: $(-, -)$ for C_3 (Cl(2), KO=6, the Connes–Chamseddine SM triple) and $(+, +)$ for K_{10} (Cl(9), KO=0, the M-theory triple [53]). The framework is CPT-invariant, with J_G providing the real structure of a spectral triple.

6.5 Jacobi identities, thermodynamics, and Einstein's equations

The Jacobi identities of the SSCA enforce, level by level (Appendix A12): the identity for (P_a, P_b, D_G) gives the Bianchi identity $\nabla_{[a} \mathcal{R}_{bc]} = 0$; the identity for (T^A, T^B, D_G) gives gauge closure $[T^A, F^B] = f^{AB}{}_C F^C$; and the identity for (P_a, T^A, D_G) gives the Yang–Mills coupling $\nabla_a F^A = [T^A, \Gamma_a \mathcal{G}_a]$.

The full Einstein–Yang–Mills system follows thermodynamically: modular equilibrium at the gap fixed point implies vanishing relative entropy across every edge, hence no heat flux through any local Rindler horizon. Jacobson's argument [54, 55] then forces $R_{\mu\nu} k^\mu k^\nu = 0$ for all null k^μ ; since local wedges cover every null direction, $R_{\mu\nu} = \Lambda_G g_{\mu\nu}$. Combined with the Yang–Mills coupling from the Jacobi hierarchy,

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda_G g_{\mu\nu} = 8\pi G_N T_{\mu\nu}^{\text{YM}}, \quad (79)$$

with G_N and Λ_G as in Theorem 6.7. A detailed proof that the thermodynamic and Jacobi routes coincide, and that no higher-curvature terms appear at the gap fixed point, is given in Appendix A12 (Jacobi identities) and Appendix A13 (Einstein equations).

6.6 KK towers and the SSCA bicomplex

Define the *derived commutator tower* $\mathcal{T}_0 = \{D_G\}$, $\mathcal{T}_{n+1} = \text{span}[\{X, O\} : X \in \{P_a, T^A\}, O \in \mathcal{T}_n]$. Because each $P_a = -i[L, X^a]$ is an $N \times N$ matrix, the Cayley–Hamilton theorem forces $\mathcal{T}_n = 0$ for $n \geq N$ (Appendix A14). Hence the tower terminates at level $N - 1 = \dim \mathcal{P}$, identifying the bulk spatial dimension as $D_{\text{bulk}} = N + 1$. This termination is an algebraic statement, independent of any continuum limit: the number of independent nested commutators of P_a and T^A with D_G cannot exceed the rank of the vertex Hilbert space.

Define the mixed curvature operator $\Xi_{aA} = [P_a, [T^A, D_G]]$. The Jacobi identity for (P_a, P_b, T^A, D_G) gives the Bianchi-type relation

$$[\mathcal{R}_{ab}, F^A] = \nabla_a \Xi_{bA} - \nabla_b \Xi_{aA}, \quad (80)$$

the discrete analogue of $R_{\mu\nu\rho\sigma} F^{\mu\nu} F^{\rho\sigma}$ couplings. This relation shows that the coupling operator Ξ_{aA} measures the failure of the Riemann curvature to commute with the gauge current; when $\Xi_{aA} = 0$ the gauge and gravity sectors decouple exactly.

Each independent cycle γ_ℓ (with holonomy angle θ_ℓ) contributes a KK tower of masses. At the homogeneous saddle, the twisted Laplacian $\Delta_{\Phi, \rho}$ has eigenvalues proportional to w_0 (the uniform edge weight). However, w_0 is dimensionless (it is an entropy), whereas a physical mass squared must have dimension of $(\text{length})^{-2}$ (or energy^2 in natural units). The spectral gap λ_1 carries the correct dimension: from $G_N = 1/(4\lambda_1)$ and $[G_N] = [\text{mass}]^{-2}$, we have $[\lambda_1] = [\text{mass}]^2$. Therefore the physical KK masses are

$$m_{(n, \ell)}^2 \approx \lambda_1 \left(n + \frac{\theta_\ell}{2\pi}\right)^2, \quad (81)$$

matching a compact circle of radius $R_\ell \propto \lambda_1^{-1/2}$ with Scherk–Schwarz twist θ_ℓ . The integer n labels the KK level. For C_3 with $\theta = \pi$, the zero mode $n = 0$ is eliminated because the twisted boundary condition is antiperiodic, and the lowest non-zero modes give masses $m_{W,Z} \sim \lambda_1$, the discrete Higgs mechanism. The spectral gap λ_1 thus plays the role of the compactification scale, and the number of independent cycles β_1 determines the number of compact dimensions.

Relation to the cycle graph Laplacian. The eigenvalues of the cycle graph Laplacian L_{C_N} are $2(1 - \cos(2\pi k/N))$, $k = 0, \dots, N - 1$; in the continuum limit $N \rightarrow \infty$ at fixed circumference $2\pi r$, these become $(k/r)^2$, the KK tower of §3.1. The KK mechanism is therefore the continuum limit of the graph Laplacian spectrum.

6.7 BTZ correlators: a three-way comparison

We now compare three independent descriptions of the thermal two-point function for the BTZ black hole: the exact CFT result, the semiclassical geodesic approximation, and the spectral holographic (SH) formula derived from the path graph P_2 .

6.7.1 Exact graph identification

The eternal BTZ black hole has two asymptotic boundaries connected by a non-traversable wormhole; its quantum state is the thermofield double

$$|\Psi\rangle = Z(\beta)^{-1/2} \sum_n e^{-\beta E_n/2} |E_n\rangle_L \otimes |E_n\rangle_R. \quad (82)$$

This structure is *exactly* the $N = 2$ path graph P_2 : two vertices (the two CFT boundaries), one edge (bipartite entanglement), no cycles (enforcing exact causal disconnection via the QMP on trees).

At the gap fixed point, the unique edge weight w^* satisfies the gap equation $w^* = S(\rho_{12})$. The graph Laplacian is

$$L_{P_2} = w^* \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad (83)$$

with eigenvalues 0 and $\lambda_1 = 2w^*$. From Theorem 6.7:

$$G_3 = \frac{1}{4\lambda_1} = \frac{1}{8w^*}, \quad \ell_G^2 = \frac{(N-1)^2}{\lambda_1} \Big|_{N=2} = \frac{1}{2w^*}. \quad (84)$$

Identifying the Bekenstein–Hawking entropy $S_{\text{BH}} = \pi r_+/(2G_3)$ with w^* yields

$$w^* = S_{\text{BH}} = \frac{\pi r_+}{2G_3}, \quad \lambda_1 = 2w^* = \frac{\pi r_+}{G_3}, \quad (85)$$

and the BTZ temperature $T = r_+/(2\pi\ell^2)$ gives $w^* = \pi^2 T^2/2$. This identification is exact; no large- N or semiclassical approximation is required.

6.7.2 The three formulae

Exact CFT. For a boundary primary of conformal dimension $\Delta = 1 + \sqrt{1 + m^2 R^2}$ (dual to a bulk scalar of mass m), the exact Euclidean two-point function on the boundary circle of circumference $\beta = 1/T$ is [4]

$$G_{\text{exact}}(t; \beta) = \left(\frac{\beta}{\pi\delta}\right)^{2\Delta} \left(2 \cosh \frac{2\pi t}{\beta} - 2\right)^{-\Delta}, \quad (86)$$

where δ is a UV regulator. For large t , $G_{\text{exact}} \sim C_\Delta e^{-2\pi\Delta t/\beta}$.

Bulk geodesic approximation. The semiclassical geodesic length connecting two boundary points at equal time separated by t is

$$\ell(t) = 2R \operatorname{arccosh}\left(\cosh \frac{2\pi t}{\beta}\right) = \frac{4\pi R t}{\beta}, \quad (87)$$

so the geodesic approximation (valid for $mR \gg 1$, as established numerically in §5.4) gives

$$G_{\text{geo}}(t; \beta) = \exp(-m\ell(t)) = \exp\left(-\frac{4\pi m R t}{\beta}\right). \quad (88)$$

For $mR \gg 1$, $\Delta \approx mR$, and both (86) and (88) decay as $e^{-2\pi\Delta t/\beta}$, confirming the geodesic approximation as the large- t , large- mR limit of the exact result.

Spectral holographic (SH) formula. In the SH framework, the two-point function between boundary vertices i, j is the massive resolvent of the graph Laplacian:

$$\langle \hat{\mathcal{O}}_i \hat{\mathcal{O}}_j \rangle = (L + m^2)_{ij}^{-1} = \int_0^\infty e^{-m^2\sigma} K_\sigma(i, j) d\sigma \quad (89)$$

For P_2 , the heat kernel $K_\sigma = e^{-\sigma L}$ has off-diagonal element $K_\sigma(1, 2) = \frac{1}{2} - \frac{1}{2}e^{-2w^*\sigma}$. The connected part, after subtracting the uniform $1/N^2 = 1/4$ contribution, is $K_\sigma^1(1, 2) = \frac{1}{2}e^{-2w^*\sigma}$. Hence

$$G_{\text{SH}}(w^*, m) = |(L + m^2)_{12}^{-1}| \quad (90)$$

$$= \int_0^\infty e^{-m^2\sigma} \frac{1}{2} e^{-2w^*\sigma} d\sigma \quad (91)$$

$$= \frac{1}{2(2w^* + m^2)}. \quad (92)$$

Substituting $w^* = \pi^2 T^2/2$, the dominant exponential decay rate is $\sqrt{2w^*} = \pi T$, reproducing the thermal Boltzmann factor $e^{-\pi T t}$ at leading order in t (after including the boundary diffusion kernel, see Remark 6.12).

The SH two-point function $(L + m^2)_{ij}^{-1} = \int_0^\infty e^{-m^2\sigma} K_\sigma(i, j) d\sigma$ becomes the geodesic approximation $e^{-m\ell}$ precisely when the heat kernel $K_\sigma(i, j)$ behaves as $e^{-d(i, j)^2/(4\sigma)}$ and the integral is evaluated by steepest descent at $\sigma^* = d(i, j)/(2m)$. The threshold $mR \gtrsim 5$ found numerically in §5.4 marks the onset of saddle-point dominance; for $mR < 5$ the full integral is required. Thus the spectral heat kernel provides an exact, non-perturbative definition of the correlator that reduces to the semiclassical geodesic approximation in the large- mR limit, explaining the numerical accuracy observed in Fig. 7.

Remark 6.12 (Position-dependence and the C_N limit). Formula (90) is the spatially averaged SH correlator on P_2 ; it cannot resolve position-dependent correlations along the boundary circle. To obtain a position-dependent result, one must use the cycle graph C_N (discretising the boundary S^1 of circumference β). Its resolvent

$$G_{\text{SH}}^{C_N}(j; w, m) = \frac{1}{N} \sum_{k=0}^{N-1} \frac{e^{2\pi i j k / N}}{2w(1 - \cos(2\pi k / N)) + m^2} \quad (93)$$

reproduces the exact CFT image sum (86) in the continuum limit $N \rightarrow \infty$, $w = N^2/\beta^2$. The P_2 formula is recovered from the $k = 1$ mode alone, which dominates for $t \gg \beta$. The accuracy hierarchy is:

Formula	Regime	Accuracy
G_{exact} (86)	all t, T, Δ	exact
G_{geo} (88)	$mR \gg 1, t \gg \beta$	$O(1/(mR))$
$G_{\text{SH}}^{C_N}$	$mR \gg 1$, any t	exact as $N \rightarrow \infty$
$G_{\text{SH}}^{P_2}$ (90)	leading order in t	exponential decay only

6.8 Toward a spectral ensemble and phase classification

The partition function over all admissible graphs is UV-finite:

$$\mathcal{Z} = \sum_N \sum_{[G] \in \mathcal{G}_N} \frac{1}{|\operatorname{Aut}(G)|} \int_{\mathcal{L}_+^G} d\mu(L) \int_{\mathcal{H}^{\beta_1}} d\Phi e^{-\operatorname{Tr} f_E(D_G/\Lambda)}, \quad (94)$$

where D_G is the graph Dirac operator, $\mathcal{L}_+^G$ is the admissible Laplacian cone, $d\mu(L)$ the Lebesgue measure on edge weights, and the holonomy integral runs over $\mathcal{H}^{\beta_1}$ (one $U(1)$ angle per independent cycle). The symmetry factors $1/|\operatorname{Aut}(G)|$ arise automatically from the groupoid C^* -algebra trace; no gauge-fixing is required.

A *trans-stratal amplitude* connects graphs with different vertex counts N or cycle structures β_1 . Such amplitudes are computed by interpolating families that cross a topology change; they are non-perturbative in the string coupling $g_s = M^{-1}$ because they change $\dim \ker D_G$, i.e. the index $\Delta \mathcal{N}_{\text{SUSY}}$. For the transition $K_{10} \rightarrow C_3$, the spectral flow equals 16, giving amplitude $\sim M^{-16}$, invisible at any finite order in g_s .

6.8.1 M-theory from the K_{10} saddle

Three algebraic arguments identify K_{10} ($N = 10$ vertices) as the M-theory saddle. Arguments 1 and 2 are unconditional; the bosonic match to BFSS at quartic order (Appendix A15) and the $\mathcal{N} = 16$ supercharge count (Theorem 6.11) are rigorous. Argument 3 below is illustrative rather than independent (entropic dominance reduces to Argument 1). Full all-orders matching to M-theory is conjectural.

1. *Clifford algebra and maximal supersymmetry.* The graph Dirac operator D_G is valued in $\text{Cl}(N-1)$. A real irreducible Majorana representation exists when $N-1 \equiv 0, 1, 2 \pmod{8}$. For $N-1 = 9$ [56],

$$\text{Cl}(9) \cong \text{Mat}_{16}(\mathbb{R}) \oplus \text{Mat}_{16}(\mathbb{R}), \quad (95)$$

and a single Majorana spinor selects one 16-dimensional block, giving $\mathcal{N}_{\text{SUSY}} = 16$ real supercharges – exactly the number for M-theory. Smaller N give fewer supercharges; larger N repeat by Bott periodicity but add no new structure. Hence K_{10} is the *minimal* graph realising maximal supersymmetry.

2. *Non-perturbative exactness of the K_{10} saddle.* For K_{10} with trivial holonomy ($\Phi_{ij} = \text{Id}$), the gauge field strength vanishes identically: $(d_\Phi \alpha)_{ij} = 0$ on every edge. The contraction parameter of the gap map is $\kappa = g_{\text{YM}}^2 K / (2\varepsilon) = 0$ because the modular spectral density K vanishes when all edge automorphisms are trivial. Therefore the gap equation $F_{ij}(\{w_{kl}\}) \equiv w_0$ has a unique fixed point $w_{ij}^* = w_0$ for *all* values of g_{YM} , with no restriction to the subcritical regime $\kappa < 1$. Thus K_{10} is non-perturbatively exact, whereas every other graph stratum requires $\kappa < 1$ for the existence and uniqueness theorem to apply.

3. *Entropic dominance (sketch).* For uniform K_N , the linear scaling of $S_E[K_N]$ in N is broken by the fluctuation determinant $Z_{1\text{-loop}} \propto \lambda_1^{-\alpha/2}$ and the Clifford-Majorana multiplicity, the latter peaking at $N=10$ by Bott periodicity. The free energy $F(N) = S_E[K_N] - \frac{1}{2} N_s(N) \log(\Lambda^2/\lambda_1)$ is minimised at $N=10$ only when Argument 1 is enforced, so this argument is not independent. Rigorous all- N minimisation remains open.

Remark 6.13 (Distinction between vertex count N and trace rank M). Two independent large parameters appear:

- **Vertex count** N : a topological label of the graph stratum; controls the bulk dimension via $D_{\text{bulk}} = 2 + (N-1) + \dim \mathcal{M}_{\text{int}}$. Summed over in the ensemble.
- **Trace rank** M : the rank of the projection in the crossed-product trace $\text{Tr}_{\text{sf}}(P) = M$; counts independently synchronised modular sectors. Physical identification: $g_s = M^{-1}$, $R_{11} = M\ell_P$.

BFSS is the $M \rightarrow \infty$ limit at *fixed* $N = 10$ [57]. The ensemble sums over N at finite M . Trans-stratal amplitudes such as $K_{10} \rightarrow C_3$ are suppressed by M^{-16} , non-perturbative in g_s and invisible at any finite order in perturbative string theory [58].

6.8.2 BFSS matrix model from the K_{10} saddle

With uniform weight w^* , $L_{K_{10}} = w^*(10I - J)$ has eigenvalues 0 (once) and $\lambda_1 = 10w^*$ (multiplicity 9). The spectral embedding gives $\mathbf{x}_i \in \mathbb{R}^9$, forming a regular 9-simplex. From Theorem 6.7:

$$G_N = \frac{1}{4\lambda_1} = \frac{1}{40w^*}, \quad (96)$$

$$\Lambda_G = -\frac{10\lambda_1}{2 \cdot 9} = -\frac{10w^*}{9}, \quad (97)$$

$$D_{\text{bulk}} = N + 1 = 11. \quad (98)$$

Introduce fluctuations $w_{ij} = w^* + M^{-1/2} \delta w_{ij}$ and define traceless $\mathfrak{su}(M)$ matrices X^I ($I = 1, \dots, 9$) by

$$\delta w_{ij} = \sqrt{10w^*} \sum_{I=1}^9 (e_I)_i (e_I)_j \text{tr } X^I, \quad (99)$$

where $\{e_I\}$ is an orthonormal basis of the λ_1 -eigenspace. The Dirac operator becomes $D_G = D_G^{(0)} + M^{-1/2} \sum_I \gamma^I \otimes [X^I, \cdot]$. Set the UV cutoff $\Lambda = M\sqrt{\lambda_1}$ and expand the spectral action $S_E = \text{Tr}_E(D_G/\Lambda)$ in a heat-kernel series. The leading terms in M^{-1} yield

$$S_E \xrightarrow{M \rightarrow \infty} \frac{1}{g_s} \int dt \text{Tr} \left(\frac{1}{2} (D_t X^I)^2 + \frac{1}{4} [X^I, X^J]^2 + \theta_\alpha \Gamma_{\alpha\beta}^I [X^I, \theta_\beta] \right), \quad (100)$$

with $g_s = \lambda_1^{-3/2}$. The X^I are the nine non-zero Laplacian eigenmodes; the 16-component Majorana fermions θ_α arise from $\ker D_G^{(0)}$. This matches the bosonic BFSS Lagrangian [37] through quartic order in X^I at leading order in M^{-1} , including the $\text{Tr}[X^I, X^J]^2$ coupling and the 16-component Majorana fermions from $\ker D_G^{(0)}$. The $g_s = M^{-1}$ identification and Planck-scale cutoff $\Lambda = M\sqrt{\lambda_1}$ give $R_{11} = M\ell_P$ and $\ell_{11} = M^{1/9}\ell_P$ as in M-theory. Higher-order matching to supergravity loops remains open.

Remark 6.14 (BFSS matrices as fuzzy spectral embedding). The $X^I \in \mathfrak{su}(M)$ are the non-commutative deformation of the λ_1 -eigenspace of $L_{K_{10}}$: classically $\mathbf{x}_i \in \mathbb{R}^9$ with commuting coordinates; quantum-mechanically $[X^I, X^J] \neq 0$ at finite M . BFSS geometry is the fuzzy 9-simplex, with non-commutativity scale $M^{-1/2}$. The matrix indices $a, b=1, \dots, M$ label modular sectors of a single vertex algebra, not extra graph vertices. N and M are algebraically distinct.

Remark 6.15 (BFSS dynamics are intra-stratal). BFSS preserves the K_{10} stratum: $M \rightarrow M \pm 1$ are D0-brane creation/annihilation, changing the observer but not the graph; $N=10$, $\beta_1=36$, and $\mathcal{N}_{\text{SUSY}}=16$ are all conserved by the BFSS Hamiltonian. The trans-stratal $K_{10} \rightarrow C_3$ amplitude lies outside BFSS by construction: it reduces $\mathcal{N}_{\text{SUSY}}$ by 16, suppressed by M^{-16} . This explains, structurally, why perturbative M-theory cannot describe the Big Bang, the formalism is intra-stratal.

Among all $N = 10$ graphs, K_{10} with trivial holonomy gives $\ker \Delta_{\Phi, \rho} \cong V_\rho$ (full representation space) and maximal $\ker \hat{R}_\mathcal{P}$ (the simplex Δ^9 has no conical singularities).

Hence $\ker D_G = V_\rho$ and $\mathcal{N}_{\text{SUSY}} = 16$. Any non-trivial holonomy or non-complete graph reduces $\mathcal{N}_{\text{SUSY}}$. Thus K_{10} is the unique maximally supersymmetric saddle of the spectral ensemble.

Remark 6.16 (Parallels with string-theoretic gauge/gravity coupling). In string theory, weak coupling $g_s \ll 1$ describes open strings (gauge fields) on D-branes; strong coupling backreacts to closed strings (gravity). In SH, the contraction parameter $\kappa = g_{\text{YM}}^2 K / (2\epsilon)$ plays the same role: $\kappa \ll 1$ decouples gauge and gravity (spectral holographic duality), $\kappa \rightarrow 1$ signals a Hagedorn transition, and $\kappa > 1$ locks them into the Einstein–Yang–Mills system.

Phase classification. The three principal strata are P_2 (BTZ/AdS₃: $N=2$, no cycles, trivial holonomy, QMP exact); C_3 (Standard Model: $N=3$, $\mathbb{Z}_2$ holonomy, $\beta_1=1$ truncates the tower, broken SUSY, massive $W^\pm, Z$); and K_{10} (M-theory: $N=10$, trivial holonomy, $D_{\text{bulk}}=11$, $\mathcal{N}_{\text{SUSY}}=16$, QMP exact). The transition $K_{10} \rightarrow C_3$ reduces SUSY from 16 to 0, releases entropy $16 \log M$ per modular sector, and is suppressed by M^{-16} , non-perturbative in $g_s=M^{-1}$, explaining why perturbative strings cannot describe the Big Bang. Ten-dimensional string vacua arise as holonomy-reduced sectors of K_{10} .

Remark 6.17 (The central test: BTZ correlator). The P_2 heat-kernel correlator reproduces the exponential thermal decay of BTZ correlators, with the spectral gap setting the temperature. Agreement in the semiclassical regime $mR \gg 1$ demonstrates that the graph heat kernel captures holographic correlators without a continuum manifold.

AdS/CFT result	Graph origin
RT formula	QMP [†] + min-cut theorem
Geodesic correlator	Saddle-point of $(L^\dagger)_{1N}$
BTZ thermal correlator	P_2 heat kernel + Gaussian diffusion
Gauge group from D-branes	Holonomy kernel: $\ker \Delta_\Phi \cong \text{Fix}(\mathcal{H})$
KK spectrum	Cycle holonomy towers
$G_N, \Lambda < 0$	λ_1 : $G_N = 1/(4\lambda_1)$, $\Lambda_G = -N\lambda_1/2(N-1)$
SUSY breaking	$\ker \hat{R}_P \cap \text{Fix}(\mathcal{H}) = \emptyset$ for C_3
Vacuum Einstein equations	Thermodynamics [†] + SSCA Jacobi identities
M-theory / BFSS	K_{10} saddle, large- M limit of spectral action

[†] Conditional on Conjecture 6.1 (QMP for $\beta_1 \geq 1$).

7 Conclusion and Outlooks

The Penrose-diagram invariant N , the number of boundary components of any conformally compactified space-time, is realised in the spectral-holography framework as the number of extremal vertices of a finite weighted

graph G . Edge entanglement weights w_{ij} and holonomies $\Phi_{ij} \in \text{Aut}(\mathfrak{g})$ then supply the bulk data the diagram alone cannot fix, and the spectral data of G generate the bulk metric, gauge fields, and SUSY content.

What the framework delivers unconditionally is, we think, the most striking part. The holonomy kernel theorem gives 9 massless gauge bosons on $C_3/\mathbb{Z}_2$, the gluons and the photon, without fine-tuning. The path graph P_2 reproduces the eternal BTZ black hole exactly, including Sen’s logarithmic correction $\alpha_{P_2}=3/2$. The complete graph K_{10} matches BFSS to quartic order at $\kappa=0$, with maximal supersymmetry forced by Bott periodicity. And the stratum invariant $\alpha=(N_\partial+1+n_{\text{mod}}-n_{\text{gh}})/2$ gives the parameter-free predictions $\alpha_{C_3}=2$ and $\alpha_{K_{10}}=19$. Conditional on the QMP (Conj. 6.1) one obtains $G_N=(4\lambda_1)^{-1}$, $\Lambda_G = -N\lambda_1/[2(N-1)]$, the RT formula, and the vacuum Einstein equations.

The numerical study of §5.4 provides a quantitative check of the geodesic–correlator duality underlying the heat-kernel form of the SH two-point function. The geodesic approximation becomes accurate for $mR \gtrsim 5$, which is precisely the steepest-descent threshold of the SH resolvent (47). This is the most concrete numerical confirmation we have that the graph heat-kernel reproduces the standard holographic correlator in the regime where the standard holographic correlator is itself reliable.

Forces appear in this picture as facets of a single algebraic non-commutativity: gravity $[P_a, D_G] \neq 0$, gauge $[T^A, D_G] \neq 0$, SUSY $\ker D_G \neq \{0\}$, CPT $J_G D_G J_G^* = \epsilon' D_G$, and the proposed loss of a global modular time at a de Sitter horizon $[\mathbf{D}, \hat{h}_i] \neq 0$. Taken as a static background, the C_3 saddle does not by itself give a universe like ours: it carries the right gauge content but sits at $\Lambda_G < 0$, with no inflation and no electroweak breaking beyond the geometric discrete-Higgs spectrum.

A more promising route lies in the algebraic structure already present inside C_3 . The graph admits an admissible $\mathbb{Z}_2$ reflection σ' ; gauging it severs the unique cycle, restores exact Markovianity on the quotient tree P_2 , and produces a codimension-two ghost vertex where the resulting end-of-the-world brane meets the bulk. By Rmk. 6.9, this ghost vertex carries a Type II₁ algebra with modular Hamiltonian automatically below threshold, so it sits permanently inside V_{isl} without tuning. The same $\mathbb{Z}_2$ holonomy that selects $\text{Fix}(\tau) = \text{su}(3) \oplus \text{u}(1)_Q$ and gives mass to $W^\pm, Z$ is localised there, so the algebraic event $[\mathbf{D}, \hat{h}_{\text{gh}}] \neq 0$ that nucleates a de Sitter patch coincides with the activation of the discrete Higgs mechanism. Electroweak symmetry breaking and de Sitter nucleation would then be two readings of one algebraic transition in the same modular sector.

This suggests a route from the unconditional C_3 results toward phenomenology. The spectral gap field Φ_G near the ghost vertex plays a dual role as order parameter for the electroweak transition and as seed for the late-time cosmological constant, with characteristic scale $v = \sqrt{\lambda_1}$ and a one-loop suppression $\Lambda_{\text{eff}} \sim \lambda_1^2/M$ from the trace rank. The relation $m_W/m_Z = \cos \theta_W$ becomes a derived statement once the gap equation is linearised around the

gauged C_3/σ' saddle, though the prefactor calculation remains to be done. Whether the same ghost-vertex data also fixes inflationary observables is left open.

The principal open problems are: (i) establishing the QMP for $\beta_1 \geq 1$ from the gap equation; (ii) whether M is genuinely dynamical or, in the CLPW sense [42], fundamentally an observer-frame choice; (iii) all-orders matching of higher BFSS supergravity loops to the spectral action; and (iv) an independent 11D computation of $\alpha_{K_{10}} = 19$. If these are answered favourably, spectral holography would constitute a background-independent, UV-complete, non-perturbative candidate for quantum gravity whose only inputs are λ_1 and M . Both, in principle, are observable. On this reading the universe we inhabit is the C_3 representation perturbed by ghost-vertex nucleation, and the $K_{10} \rightarrow C_3$ entropy release, suppressed by M^{-16} and irreversible by the second law, is what locks the cosmology into that representation.

Supplementary Proofs

All proofs are self-contained. Definitions of the graph, edge weights, holonomies, Laplacian, Dirac operator, etc., are as given in the main text; they are reproduced here only when necessary for clarity.

A1. Tree independence of edge entropies

Lemma .1. *Let $G = (V, E)$ be a finite connected graph, and let T, T' be two spanning trees. For each edge $\{i, j\} \in E$, the reduced density matrices $\rho_{ij}^{(T)}$ and $\rho_{ij}^{(T')}$ produced by the spanning-tree construction with fixed Schmidt spectra $\{\lambda_\alpha^{(ij)}\}$ are unitarily equivalent. In particular $S(\rho_{ij})$ does not depend on the tree.*

Proof. The Hilbert space at each vertex is separable but infinite-dimensional [59], so we work with a spectral cutoff $P_\Lambda^{(i)} = \mathbf{1}_{|\hat{h}_i| \leq \Lambda}$ and the finite-dimensional restriction $\mathcal{H}_i^\Lambda = P_\Lambda^{(i)} \mathcal{H}_i$. On these truncated spaces, the entanglement unitaries U_{ij}^Λ still produce bipartite states whose Schmidt spectra [60] are the prescribed $\{\lambda_\alpha^{(ij)}\}$.

The key fact is that two spanning trees of a finite graph produce states that differ only by local gauge transformations at each vertex, this is the standard MPS-on-a-tree gauge-freedom lemma [48]. So for every Λ there is a vertex-local unitary $U_{TT'}^\Lambda$ with

$$U_{TT'}^\Lambda \rho_{ij}^{(T), \Lambda} (U_{TT'}^\Lambda)^\dagger = \rho_{ij}^{(T'), \Lambda}.$$

We need to send $\Lambda \rightarrow \infty$. Because the Schmidt coefficients $\lambda_\alpha^{(ij)}$ decay super-polynomially (otherwise w_{ij} would diverge), the truncated states converge to the full ones in trace norm [59]. Extending each $U_{TT'}^\Lambda$ by the identity on the orthogonal complement gives a sequence in the unit ball of $\mathcal{B}(\mathcal{H}_i \otimes \mathcal{H}_j)$, which is weak-operator compact by Banach–Alaoglu [59]. A subnet converges weakly to some $U_{TT'}$; weak limits of isometries and co-isometries remain isometries and co-isometries [61], so the limit is unitary. Passing to the limit in the conjugation identity

gives $U_{TT'} \rho_{ij}^{(T)} U_{TT'}^\dagger = \rho_{ij}^{(T')}$, and entropy is invariant under unitary conjugation [62, 63]. $\square$

A2. Unique gap fixed point

Theorem .2. *Let $\kappa = g_{\text{YM}}^2 K / (2\varepsilon)$ with $K = \|\rho_0\|_{L^1}$ and $\varepsilon > 0$ a uniform lower bound on the weights (i.e., $w_{kl} \geq \varepsilon$ for all edges). Assume $\kappa < 1$. Then the gap map $F : [\varepsilon, w_0]^{|\mathcal{E}|} \rightarrow [\varepsilon, w_0]^{|\mathcal{E}|}$ defined by $F_{ij}(\{w_{kl}\}) = S(\rho_{i \cup j}(\{w_{kl}\}))$ satisfies $\|DF\|_{\ell^\infty \rightarrow \ell^\infty} \leq \kappa$ and has a unique fixed point $\{w_{ij}^*\} \subset \mathcal{L}_+^G$, depending smoothly on g_{YM} .*

Proof. Using the Araki representation of the relative modular operator [63, 64], the entropy can be written as

$$S(\rho_{i \cup j}) = - \int_0^\infty \frac{d\lambda}{\lambda} \text{Tr}[\rho_{ij}(\log \lambda) \Delta_{\omega_i | \omega_j}] + \text{const},$$

and the twisted Laplacian gives the explicit formula

$$F_{ij}(\{w_{kl}\}) = w_0 - \frac{g_{\text{YM}}^2}{2} \int_0^\infty \frac{d\lambda}{\lambda} \rho_{ij}(\log \lambda; \{w_{kl}\}) \text{Tr}[(d_\Phi \alpha)_{ij}^2].$$

Differentiating with respect to w_{kl} yields

$$\frac{\partial F_{ij}}{\partial w_{kl}} = - \frac{g_{\text{YM}}^2}{2} \int_0^\infty \frac{d\lambda}{\lambda} \frac{\partial \rho_{ij}}{\partial w_{kl}}(\log \lambda; \{w_{kl}\}) \text{Tr}[(d_\Phi \alpha)_{ij}^2].$$

The spectral density ρ_{ij} is L^1 [59] and its variation satisfies

$$\left\| \frac{\partial \rho_{ij}}{\partial w_{kl}} \right\|_{L^1} \leq \frac{\|\rho_0\|_{L^1}}{\varepsilon},$$

because the weights are bounded below by ε , so a unit change in a weight can change the modular spectral density by at most $1/\varepsilon$ times the reference density. Moreover $|\text{Tr}[(d_\Phi \alpha)_{ij}^2]| \leq 1$ by normalisation of the gauge field strength [65]. Hence

$$\left| \frac{\partial F_{ij}}{\partial w_{kl}} \right| \leq \frac{g_{\text{YM}}^2}{2} \cdot \frac{K}{\varepsilon} = \kappa,$$

where $K = \|\rho_0\|_{L^1}$. Thus $\|DF\|_{\ell^\infty \rightarrow \ell^\infty} \leq \kappa < 1$.

The set $[\varepsilon, w_0]^{|\mathcal{E}|}$ is compact and convex, and F maps it into itself because the entropy is bounded between 0 and $\log \dim \mathcal{H}_i$ [60, 63]. By the Banach fixed-point theorem, F has a unique fixed point w^* . For smooth dependence on g_{YM} , consider $G(w, g_{\text{YM}}) = w - F(w, g_{\text{YM}})$. The derivative $\partial G / \partial w = I - DF$ is invertible because $\|DF\| < 1$. The derivative $\partial G / \partial g_{\text{YM}}$ exists because F is differentiable under the integral sign (dominated convergence [59], as ρ_{ij} is uniformly bounded). The implicit function theorem then gives smooth dependence of $w^*(g_{\text{YM}})$. $\square$

A3. Exact QMP on trees

Lemma .3. *If G is a tree, then for every tripartition $A, B, C \subset V$ with B separating A from C , $I(A : C|B) = 0$.*

Proof. Induction on $|V|$. The base case is a single edge, where any separating tripartition either puts C empty (so

the conditional mutual information is trivially zero [60] or symmetrically A empty.

For the inductive step, pick a leaf v with unique neighbour u . There are two cases according to where v sits:

v lies on one side of the partition. Without loss of generality $v \in A$. Remove v to get a smaller tree G' with $A' = A \setminus \{v\}$, $B' = B$, $C' = C$. Separation is preserved in G' because the only edge incident to v went through u , and u was already in B (otherwise v would have provided a path from A to C bypassing B). The induction hypothesis gives $I(A' : C'|B') = 0$ in G' . The fixed-Schmidt construction makes the reduced state on $V \setminus \{v\}$ identical whether v was originally present or not, since the bipartite state on the edge $\{u, v\}$ is in canonical form [60]. Hence $I(A : C|B) = I(A' : C'|B') = 0$.

$v \in B$. Then v is part of the separating set. The fixed-Schmidt state on the leaf edge factorises after conditioning on u , so v contributes no additional conditional correlation. Restricting attention to $V \setminus \{v\}$ and keeping $u \in B$ reduces to the previous case (or to the base case if $|V| = 2$). This is the standard quantum Markov chain argument of [66]; we omit the routine chain-rule bookkeeping. $\square$

A4. Cocycle trivialisation at the gap fixed point

Proposition .4. *At the gap fixed point $w_{ij}^* = S(\rho_{i \cup j})$, the Connes–Radon–Nikodym cocycle satisfies $(D\omega_i : D\omega_j)_t = 1$ for all $t \in \mathbb{R}$ and every edge.*

Proof. The gap equation says that the entropy of the bipartite reduced state equals the weight that defines it — in other words, the states ω_i and ω_j assign the same expectation value to the modular Hamiltonian of the edge algebra $\mathcal{M}_{ij}$: $\omega_{ij}(\log \Delta_{\omega_i|_{\omega_j}}) = 0$. Both states are faithful on $\mathcal{M}_{ij}$ because the vertex algebras are Type II $_{\infty}$ after the crossed product [41, 44]. Araki’s theorem [63, 64] says that vanishing relative entropy between two faithful normal states forces them equal: $\omega_i|_{\mathcal{M}_{ij}} = \omega_j|_{\mathcal{M}_{ij}}$. Equal states have unit relative modular operator, $\Delta_{\omega_i|_{\omega_j}} = \mathbf{1}$, and the cocycle is the imaginary power of this operator, so it is the identity for all t . $\square$

A5. Holonomy kernel theorem

Theorem .5. *Let G be a connected graph with holonomy group $\mathcal{H} \subseteq \text{Aut}(\mathfrak{g})$. For any representation (ρ, V) , define the twisted graph Laplacian $\Delta_{\Phi, \rho}$ by*

$$(\Delta_{\Phi, \rho} \alpha)_i = \sum_{j \sim i} w_{ij} (\alpha_i - \rho(\Phi_{ij}) \alpha_j).$$

Then $\ker \Delta_{\Phi, \rho} \cong \text{Fix}(\mathcal{H}|_{\rho}) := \{v \in V : \rho(P)v = v \ \forall P \in \mathcal{H}\}$, and the dimension of the kernel is independent of all edge weights $\{w_{ij}\}$.

Proof. Fix a spanning tree T and a root vertex v_0 . For any $v \in \text{Fix}(\mathcal{H}|_{\rho})$, define $\alpha \in V^N$ by $\alpha_{v_0} = v$ and for any other vertex i , let $\alpha_i = \rho(P_{v_0 \rightarrow i}^T)v$, where $P_{v_0 \rightarrow i}^T$ is the product of Φ ’s along the unique path in T . For each non-tree edge $\{i, j\} \notin T$, consider the cycle γ from v_0 to

i along T , then along $i \rightarrow j$, then back to v_0 along T . Its holonomy $P_{\gamma} \in \mathcal{H}$ satisfies $\rho(P_{\gamma})v = v$ because v is fixed by $\mathcal{H}$. This implies $\alpha_i = \rho(\Phi_{ij})\alpha_j$, so α is a parallel section and therefore $\alpha \in \ker \Delta_{\Phi, \rho}$. The map $\Psi(v) = \alpha$ is injective because $\Psi(v)_{v_0} = v$.

Conversely, let $\alpha \in \ker \Delta_{\Phi, \rho}$. Set $v = \alpha_{v_0}$. For each vertex i , the parallel-section condition along tree edges forces $\alpha_i = \rho(P_{v_0 \rightarrow i}^T)v$ (otherwise the Laplacian would be non-zero [49]). For each non-tree edge $\{i, j\}$, the condition $\alpha_i = \rho(\Phi_{ij})\alpha_j$ gives $\rho(P_{\gamma})v = v$ for the associated cycle γ . Since cycles generate $\mathcal{H}$ [67], we have $\rho(P)v = v$ for all $P \in \mathcal{H}$, i.e., $v \in \text{Fix}(\mathcal{H}|_{\rho})$. Thus $\alpha = \Psi(v)$, proving surjectivity.

The construction never uses the edge weights w_{ij} , so the kernel dimension is weight-independent. $\square$

A6. Gravitational constants from the spectral gap

Theorem .6. *At the gap fixed point of a connected graph with $N \geq 2$ vertices, $G_N = 1/(4\lambda_1)$ and $\Lambda_G = -N\lambda_1/[2(N-1)]$.*

Proof. Newton’s constant. Start with the Connes–Chamseddine heat-kernel expansion of the spectral action [18, 53, 68]:

$$S_E = \frac{f_2 \Lambda^2}{(4\pi)^{d/2}} \int_{\mathcal{P}} R + O(\Lambda^0), \quad f_2 = \int_0^{\infty} u f_E(u) du.$$

Match this to $\frac{1}{16\pi G_N} \int R \sqrt{g} d^d x$ [69] at the self-gravity scale $\Lambda^2 = 2\lambda_1$. The dimensional coefficients track through cleanly because $\mathcal{P}$ is piecewise flat and convex with Euler characteristic 1; Gauss–Bonnet gives $\int_{\mathcal{P}} R d\text{vol} = 4\pi$. The calculation is exactly the one carried out for the standard model spectral triple in [53] and gives $G_N = 1/(4\lambda_1)$. We do not reproduce it here.

Cosmological constant. For a homogeneous saddle, $\text{Tr}(L^\dagger) = (N-1)/\lambda_1$ [49], and the spectral embedding places vertices on a regular simplex with squared edge length $2/(N-1)$. Matching the angular polytope’s boundary metric to the standard AdS boundary at radius ℓ_G gives $\ell_G^2 = (N-1)^2/\lambda_1$. The cosmological constant of an AdS_D space of curvature radius ℓ_G is $\Lambda = -(D-1)(D-2)/(2\ell_G^2)$ [6, 69]. With $D = N+1$, this becomes

$$\Lambda_G = -\frac{N(N-1)}{2\ell_G^2} = -\frac{N\lambda_1}{2(N-1)},$$

which is the claimed expression. $\square$

A7. Metric-mode rigidity

Proof of Theorem 6.10. The Lichnerowicz formula [70] on the angular polytope reads $\square_{\mathcal{P}} h_{ab} + R_{\mathcal{P}}^{cd} h_{cd} = S_{ab}$, with h_{ab} the metric perturbation and S_{ab} the matter source. On the interior of every simplex face of $\mathcal{P}$ the Riemann tensor vanishes (piecewise flatness [71, 72]), so $R_{\mathcal{P}}^{cd} \equiv 0$ there. The angular Laplacian $\Delta_{\mathcal{P}}$ on a piecewise-flat polytope has discrete spectrum $\{\mu_a\}_{a \geq 0}$ with $\mu_0 = 0$ [72]; the nonzero modes Y_a exist as distributions supported on the

singular strata but produce zero source for the linearised Einstein tensor on the bulk interior [69]. With asymptotic boundary condition $\alpha_a(\infty) = 0$, the radial ODE $\alpha''_a + (n/r)\alpha'_a - (\mu_a/r^2)\alpha_a = 0$ admits only the trivial solution. The scalar mode $a = 0$ survives because the mass term $\delta\mu r^{1-n}$ carries no angular structure and lives on the bulk warped product, not on $\mathcal{P}$. In the continuum limit $\mathcal{P}_k \rightarrow (\mathcal{M}_\infty, \gamma^{(0)})$ with $d = \lim(N_k - 1)$, the angular curvature $R_{\mathcal{P}}^{cd}{}_{ab}$ becomes non-zero on a positive-volume set, providing non-trivial source for every eigenmode of $\Delta_{\mathcal{M}_\infty}$. Weyl's law for the Laplacian on a smooth compact d -manifold then gives the stated counting. $\square$

A8. Proof of the discrete island formula

We prove Proposition 6.8. Write $|\gamma|_w = \sum_{\{i,j\} \in \gamma} \bar{\Phi}_G(i, j)$ for the total weight of an edge cut, with $\bar{\Phi}_G(i, j) = \frac{1}{2}(\Phi_G(i) + \Phi_G(j))$.

Lemma .7. *The unrestricted optimum $I^* = \arg \min_{I \subseteq V \setminus V_\partial} |\gamma_{A \cup I}^{\min}|_w$ lies entirely inside V_{isl} .*

Proof. Suppose for contradiction that I^* contains some $i_0 \notin V_{\text{isl}}$. Then $\Phi_G(i_0) \geq \lambda_1^{\text{thr}}$, which forces every edge incident to i_0 to have weight $\bar{\Phi}_G(i_0, j) \geq \frac{1}{2}\lambda_1^{\text{thr}} > 0$. Now consider the modified island $I' = I^* \setminus \{i_0\}$. The minimal cut for $A \cup I'$ no longer has to encircle i_0 , so it can pass through it instead. By the standard max-flow/min-cut exchange argument applied to weighted graphs [73], passing through i_0 strictly beats encircling it whenever all incident edges have positive weight. So $|\gamma_{A \cup I'}^{\min}|_w < |\gamma_{A \cup I^*}^{\min}|_w$, contradicting the optimality of I^* . $\square$

Proof of Proposition 6.8. Upper bound. For any $I \subseteq V_{\text{isl}}$, strong subadditivity [62, 63] together with purity of the global graph state gives $S(A) \leq S(A \cup I)$. Applying the graph RT formula (Theorem 6.7; [25, 74]) to the enlarged region identifies the right-hand side with $(4G_N)^{-1} |\gamma_{A \cup I}^{\min}|_w$. The wedge hypotheses of [44, 45] apply to each $i \in V_{\text{isl}}$ because the two-sided modular spectrum (Appendix A4) makes the local algebra properly infinite. Minimising over I gives the bound.

Lower bound. Lemma .7 restricts the optimum to V_{isl} , and the quantum extremal surface prescription of [28, 29, 75, 76] saturates it at $I = I^*$. When $V_{\text{isl}} = \emptyset$ only $I = \emptyset$ is admissible and the formula reduces to standard RT [25]. $\square$

A9. Stratum invariant α

We derive $\alpha = \frac{1}{2}(N_\partial + 1 + n_{\text{mod}} - n_{\text{gh}})$.

Around a gap saddle (w^*, Φ^*) , the spectral-ensemble partition function splits as $\mathcal{Z} \supset e^{-S_{\text{spec}}^{(0)}} \cdot Z_{1\text{-loop}}$, and the one-loop piece is a product of Gaussian determinants [77]: one for metric moduli, one for holonomies, and one for the Faddeev–Popov ghosts [51, 78] that fix the residual gauge redundancies. The strategy is to track the λ_1 -scaling of each factor; the rest is bookkeeping.

Metric moduli. By Theorem 6.10, every trace-constrained weight perturbation reduces at finite N to a

single scalar mode plus the N_∂ asymptotic charges $\{q_a\}$. One of those charges is killed by the trace constraint $\sum q_a = -\delta\mu$, leaving $N_\partial + 1$ physical moduli in total (the asymptotic charges plus the global mass shift). Dimensional analysis of the spectral action, the same scaling that gives $G_N \propto \lambda_1^{-1}$ in [18], forces the Hessian on this moduli space to scale linearly in λ_1 , so a Gaussian integral over $N_\partial + 1$ variables gives $\lambda_1^{-(N_\partial+1)/2}$.

Holonomy moduli. For each independent cycle γ with *continuous* gauge group ($U(1)$ in our applications), the modulus is the phase $\theta_\gamma \in [0, 2\pi)$. There are $n_{\text{mod}} = \beta_1(G)$ of these [67]. The Hessian again scales as λ_1 for the same dimensional reason, giving $\lambda_1^{-n_{\text{mod}}/2}$. *Discrete* holonomy (the $\mathbb{Z}_2$ on C_3 , for example) contributes a finite sum rather than an integral, with no λ_1 scaling, this is why $n_{\text{mod}} = 0$ for C_3 in the table.

Ghosts. The N_∂ asymptotic boundaries carry diffeomorphism redundancies that must be gauge-fixed. The standard BRST argument [51] requires $n_{\text{gh}} = N_\partial - 1$ Faddeev–Popov ghosts [78] for continuous holonomy, and zero ghosts for discrete holonomy. Each ghost is anti-commuting, so its Gaussian determinant contributes $\lambda_1^{+1/2}$ (the opposite sign to a bosonic mode) [77]. The combined ghost factor is $\lambda_1^{+n_{\text{gh}}/2}$.

Combining.

$$Z_{1\text{-loop}} \propto \lambda_1^{-(N_\partial+1)/2 - n_{\text{mod}}/2 + n_{\text{gh}}/2} = \lambda_1^{-\alpha},$$

with $2\alpha = N_\partial + 1 + n_{\text{mod}} - n_{\text{gh}}$ as claimed. From Theorem 6.7, $S_{\text{BH}} \propto \lambda_1^{-1}$ [79, 80], so $\log Z_{1\text{-loop}} = \alpha \log \lambda_1 \sim -\alpha \log S_{\text{BH}}$, giving the logarithmic entropy correction $\Delta S = \alpha \log S_{\text{BH}} + O(1)$. The result is a stratum invariant because N_∂ , β_1 and n_{gh} are topological data of the graph [67], the same universality that underpins Sen's logarithmic correction analysis [52].

A10. Brane stability under perturbations

Proof. Domain walls. Let $\sigma \in \text{Aut}(G)$ be an admissible reflection: $\sigma^2 = \text{Id}$, $\det \sigma = -1$, and it preserves edge weights and holonomies. Write a weight perturbation as $\delta w_{ij} = \delta w_{ij}^+ + \delta w_{ij}^-$ with

$$\delta w_{ij}^\pm = \frac{1}{2}(\delta w_{ij} \pm \delta w_{\sigma(i)\sigma(j)}).$$

The Connes–Radon–Nikodym cocycle $(D\omega_\epsilon : D\omega_\epsilon \circ \sigma)_t$ is trivial iff $\omega_\epsilon = \omega_\epsilon \circ \sigma$, which holds exactly when $w_{ij} = w_{\sigma(i)\sigma(j)}$, i.e., $\delta w^- = 0$. If $\delta w^- \neq 0$, then for any $\epsilon > 0$ no faithful normal conditional expectation [41] from $\mathcal{M}_G$ onto the fixed-point subalgebra $\mathcal{M}_G^+$ exists, and the wall dissolves discontinuously at $\epsilon = 0^+$.

Conical branes. A $\mathbb{Z}_k$ rotation that fixes an interior point $p_0 \in \mathcal{P}$ determines a deficit angle $\delta = 2\pi(1 - 1/k)$ and a tension $T_0 = (k - 1)\lambda_1/k$. Both depend only on k and the spectral gap λ_1 , which is invariant under any $\mathbb{Z}_k$ -equivariant weight perturbation. Hence the brane persists with continuously varying tension. If the perturbation breaks the $\mathbb{Z}_k$ symmetry, the fixed point is no longer a conical singularity; the curvature becomes smooth there and the brane disappears. $\square$

A11. Graph Weitzenböck identity

Proof. Let $\{x_i^a\}_{a=1}^{N-1}$ be the spectral embedding coordinates. For an edge $\{i, j\}$, define the unit tangent vector $e_{ij}^\mu = (x_j^\mu - x_i^\mu)/|x_j - x_i|$ and $\gamma_{ij} = \sum_\mu e_{ij}^\mu \gamma^\mu$, where γ^μ generate $\text{Cl}(N-1)$ with $\{\gamma^\mu, \gamma^\nu\} = 2\delta^{\mu\nu}$ [56, 81]. The graph Dirac operator is

$$(D_G \psi)_i = \sum_{j \sim i} \sqrt{w_{ij}} \gamma_{ij} U_{ij} \psi_j,$$

with U_{ij} the parallel transport [65]. In a local orthonormal frame $\{e_a\}$ on a simplex of $\mathcal{P}$, we can write $D_G = \sum_a \gamma^a \nabla_a$, where ∇_a is the covariant derivative induced by the graph connection [81]. Squaring [70],

$$D_G^2 = \sum_a \nabla_a^2 + \frac{1}{2} \sum_{a < b} \gamma^{ab} [\nabla_a, \nabla_b].$$

The first term is the twisted Laplacian $\Delta_{\Phi, \rho}$. The commutator $[\nabla_a, \nabla_b]$ is the curvature of the connection [65]. Its grade-2 component splits into two contributions:

- A term supported on the singular strata of $\mathcal{P}$ (vertices, edges, etc.) giving the distributional scalar curvature $\frac{1}{4} \widehat{R}_{\mathcal{P}}$ [71]. The factor $\frac{1}{4}$ arises from the Clifford algebra identity $\frac{1}{2} \sum_{a < b} \gamma^{ab} R_{abcd} \gamma^{cd} = \frac{1}{4} R$ in the continuum (see [56, 81]), and on a piecewise-flat polytope the same factor holds because the curvature is concentrated at conical singularities; the angular deficit at a vertex is $2\pi - \sum_F \theta_F$ [71], and the contribution to $\widehat{R}_{\mathcal{P}}$ is $2(2\pi - \sum_F \theta_F)$, yielding the factor $\frac{1}{4}$ after the Clifford contraction.
- A term supported on non-contractible cycles giving the holonomy flux $F_\Phi = \sum_\gamma \rho(P_\gamma - \text{Id})$, where the sum runs over independent cycles [67]. This term also has grade 2, and its coefficient is determined by the parallel transport around the cycle; when $\rho(P_\gamma) = \text{Id}$, the contribution vanishes.

The three terms are orthogonal because they act on different Clifford grades (∇_a^2 is grade 0, curvature terms are grade 2 [56]) and have disjoint support (the singular strata are vertices and edges, while cycles are topological and do not intersect the singularities). Hence

$$D_G^2 = \Delta_{\Phi, \rho} + \frac{1}{4} \widehat{R}_{\mathcal{P}} + F_\Phi.$$

□

A12. Jacobi hierarchy of the SSCA

Proof. First, compute the commutator of the momentum operators. Let X^a be the diagonal matrix with entries x_i^a . Then $P_a = -i[L, X^a]$. Using the Jacobi identity,

$$[P_a, P_b] = -[[L, X^a], X^b] + [[L, X^b], X^a].$$

Since X^a and X^b are diagonal, they commute: $[X^a, X^b] = 0$. Therefore

$$\begin{aligned} [[L, X^a], X^b] &= [L, [X^a, X^b]] + [X^a, [L, X^b]] \\ &= 0 + [X^a, [L, X^b]] \\ &= -[X^a, P_b]. \end{aligned}$$

Similarly, $[[L, X^b], X^a] = -[X^b, P_a]$. Thus $[P_a, P_b] = [X^a, P_b] - [X^b, P_a]$. This is the discrete torsion tensor. In the continuum limit, this expression becomes the Riemann curvature $\mathcal{R}_{ab}$ [69]. For the finite graph, we define $\mathcal{R}_{ab}$ by this commutator; it is a matrix whose entries are supported on conical singularities [71]. The remaining commutation relations are:

$$[P_a, D_G] = \Gamma_a \mathcal{G}_a,$$

$$[T^A, D_G] = F^A$$

$$\{D_G, D_G^\dagger\} = 2\Delta_{\Phi, \rho} + 2\mathcal{R}_P.$$

Now verify the three Jacobi identities:

Bianchi identity. $\text{Jac}(P_a, P_b, D_G) = [[P_a, P_b], D_G] + [[P_b, D_G], P_a] + [[D_G, P_a], P_b] = 0$. Substituting $[P_a, P_b] = \mathcal{R}_{ab}$ and $[P_a, D_G] = \Gamma_a \mathcal{G}_a$ gives

$$[\mathcal{R}_{ab}, D_G] = \nabla_a (\Gamma_b \mathcal{G}_b) - \nabla_b (\Gamma_a \mathcal{G}_a).$$

Antisymmetrising over a, b, c yields $\nabla_{[a} \mathcal{R}_{bc]} = 0$, the Bianchi identity [69].

Gauge closure. $\text{Jac}(T^A, T^B, D_G) = [[T^A, T^B], D_G] + [[T^B, D_G], T^A] + [[D_G, T^A], T^B] = 0$. Using $[T^A, T^B] = f^{AB}{}_C T^C$ and $[T^A, D_G] = F^A$, we obtain $[T^A, F^B] = f^{AB}{}_C F^C$, the Lie algebra consistency condition.

Yang–Mills coupling. $\text{Jac}(P_a, T^A, D_G) = [[P_a, T^A], D_G] + [[T^A, D_G], P_a] + [[D_G, P_a], T^A] = 0$. Since $[P_a, T^A] = 0$, this becomes $[F^A, P_a] = [T^A, \Gamma_a \mathcal{G}_a]$. With $\nabla_a F^A = [P_a, F^A]$ [65], we have $\nabla_a F^A = [T^A, \Gamma_a \mathcal{G}_a]$, the discrete Yang–Mills equation. □

A13. Thermodynamic derivation of the vacuum Einstein equations

Proof. This derivation assumes the Quantum Markov Property (Conj. 6.1) for graphs with cycles, which guarantees the existence of a global timelike Killing vector [69] (proved unconditionally on trees).

Take any bulk point p . By wedge coverage (Voronoi cells of $\mathcal{P}$ cover the spatial slice), there exists a vertex i such that p lies in the interior of the causal wedge W_i associated to $\widehat{\mathcal{M}}_i$ [7, 43]. Let $\mathcal{H}_p$ be a local Rindler horizon generated by the boost K_i (the modular Hamiltonian of $\widehat{\mathcal{M}}_i$) [43, 82]. The Unruh temperature is $T_U = 1/(2\pi)$ [82] (from the crossed-product normalisation $\beta = 2\pi$), and the Bekenstein–Hawking entropy is $S = \mathcal{A}/(4G_N)$ [79, 80].

At the gap fixed point, the relative entropy between adjacent vertex algebras vanishes [64], giving $\delta S = 0$ across $\mathcal{H}_p$. The first law $\delta Q = T_U \delta S$ [83] then forces $\delta Q = 0$, i.e., no heat flux crosses the horizon.

On a stationary Killing horizon, the expansion θ and shear $\sigma_{\mu\nu}$ vanish [69]. The Raychaudhuri equation [84] for null geodesics with affine parameter λ is

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} - R_{\mu\nu}k^\mu k^\nu.$$

Since $\theta = 0$ and $\delta Q = 0$ implies $d\theta/d\lambda = 0$ (otherwise the horizon would expand or contract, changing entropy [54]), we obtain $R_{\mu\nu}k^\mu k^\nu = 0$ for every null generator k^μ of $\mathcal{H}_p$.

Because the collection of all local Rindler horizons covers every null direction at p [54, 55], the condition $R_{\mu\nu}k^\mu k^\nu = 0$ holds for all null k^μ at p . A symmetric tensor with this property is proportional to the metric: $R_{\mu\nu} = \Lambda g_{\mu\nu}$ [69]. (Lemma: If $T_{\mu\nu}k^\mu k^\nu = 0$ for all null k^μ , then $T_{\mu\nu} = \lambda g_{\mu\nu}$ with $\lambda = T/(d)$ where $T = g^{\mu\nu}T_{\mu\nu}$ [10].) The warped product structure of the bulk (forced by the existence of a global timelike Killing vector) and the piecewise flatness of $\mathcal{P}$ then fix $\Lambda = \Lambda_G$ as in Appendix A6. Hence $G_{\mu\nu} + \Lambda_G g_{\mu\nu} = 0$ holds at every point. $\square$

A14. Tower termination at level $N - 1$

Proof. Each $P_a = -i[L, X^a]$ is an $N \times N$ matrix, with off-diagonal entries proportional to the embedding coordinate differences $x_i^a - x_j^a$. A nested commutator $[P_{a_1}, [P_{a_2}, [\dots, [P_{a_n}, D_G] \dots]]$ is, by multilinearity, a sum of terms each carrying a wedge product of n such coordinate differences, viewed as elements of $(\mathbb{R}^{N-1})^*$. The space $\bigwedge^n(\mathbb{R}^{N-1})^*$ has dimension zero whenever $n \geq N$, so the nested commutator vanishes identically at that level. Hence the tower terminates, and the bulk spatial dimension, counted by the depth of non-trivial nested commutators, is exactly $N - 1$. $\square$

A15. BFSS matrix model from the K_{10} saddle

Proof. K_{10} has $N = 10$, trivial holonomy, and $N - 1 = 9$. $\text{Cl}(9)$ admits a Majorana representation of dimension 16 [56, 85], giving the maximal $\mathcal{N}_{\text{SUSY}} = 16$. At the uniform saddle, $L_0 = w^*(10I - J)$ has eigenvalues 0 (once) and $\lambda_1 = 10w^*$ (multiplicity 9) [49]; fix an orthonormal basis $\{e_I\}_{I=1}^9$ of the λ_1 -eigenspace.

The crossed-product trace at each vertex carries rank $\text{Tr}_{\text{sf}}(P) = M$ counting synchronised modular sectors [44, 45], so $\text{Tr} = M \cdot \text{tr}$ where tr ranges over $\mathbb{C}^{10} \otimes \text{Cl}(9) \otimes V_\rho$. Introduce fluctuations $w_{ij} = w^* + M^{-1/2}\delta w_{ij}$ with

$$\delta w_{ij} = \sqrt{10w^*} \sum_{I=1}^9 (e_I)_i (e_I)_j \text{tr} X^I,$$

X^I traceless $M \times M$ Hermitian [85]. Setting $\Lambda = M\sqrt{\lambda_1}$ (forced by the self-gravity condition combined with $g_s = M^{-1}$) [57], the spectral action $S_E = \text{Tr} f_E(D_G/\Lambda)$ expands in heat-kernel moments $a_k = \int_0^\infty \sigma^{k-1} f_E(\sigma) d\sigma$ [68].

Working through the $k = 0, 1, 2$ terms as in [53], the $k \leq 1$ contributions give field-independent constants; the dynamical part appears at $k = 2$. The cross term $\text{tr}(L_0 A)$ vanishes because A is traceless and L_0 is orthogonal to the all-ones direction [49], leaving

$$S_E \supset a_2 \Lambda^{-4} \cdot 10w^* \cdot \dim \text{Cl}(9) \cdot \frac{1}{2} \sum_{I,J} \text{tr}([X^I, X^J]^2).$$

The kinetic term $\frac{1}{2}(D_t X^I)^2$ comes from the same expansion with time-dependent fluctuations and the gauge field A_t coupling to time derivatives [85]. After absorbing constants into a field rescaling $Y^I = \alpha X^I$ and identifying

$g_s = M^{-1}$, the resulting action is precisely

$$S_E = \frac{1}{g_s} \int dt \text{Tr} \left(\frac{1}{2} (D_t Y^I)^2 + \frac{1}{4} [Y^I, Y^J]^2 \right) + \text{ferm.},$$

the BFSS action [37, 38]. The 16 Majorana fermions θ_α arise from $\ker D_G^{(0)}$ [56] and couple via $[\theta_\alpha, X^I]$ as in the standard derivation [37].

A transition $K_{10} \rightarrow C_3$ changes $\dim \ker D_G$ by $\Delta \mathcal{N}_{\text{SUSY}} = 16$. The spectral-flow index theorem on the graph cylinder [58] then gives amplitude $\sim M^{-16} = g_s^{16}$, non-perturbative in the string coupling and invisible at any finite order of perturbative string theory. $\square$

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